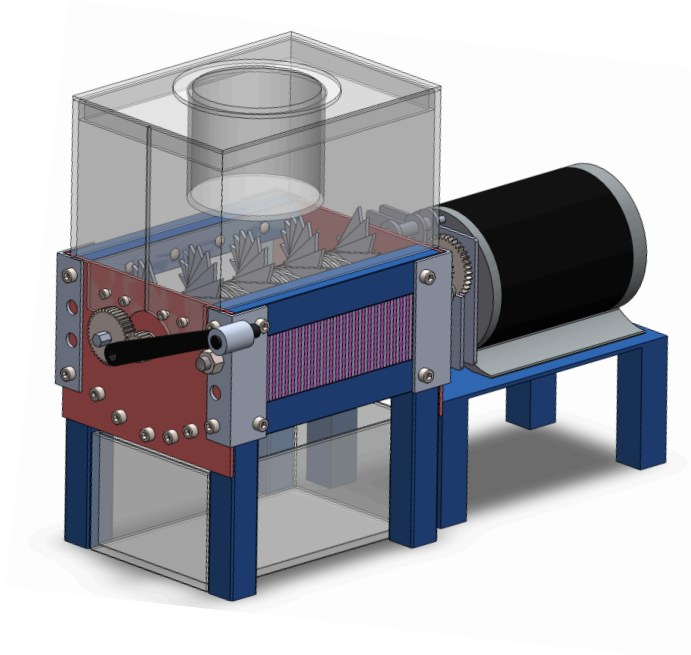




MONASH University
Engineering

Final Report of MEC3416 Machine Design

Final Machine Design Report: Structural Analysis, Component Selection, and Assembly for a Sustainable Shredder



Group 305

James Robinson -

November 3rd, 2025

1 Executive Summary

In this project we aimed to design a plastic shredder that incorporates safety, cost effectiveness and ergonomics to give the best user experience. The shredder is designed for domestic use in homes by people interested in reducing their plastic waste and having a positive impact on the environment.

The design incorporates many features, the shredding part is a single octagonal shaft with 3mm blades that mesh with a row of teeth which the plastics are crushed against. Each blade is offset by 15° making a wave action by which the shredder acts to improve the cutting performance, ensure parts are reinserted properly and reduce the instances of jamming. The handle provides the manual option and can attach directly to the shaft for direct input or through the geared input that provides a 2 to 1 torque reduction, for breaking through jams and providing more inputting torque. The electric version uses a small motor with a gearbox that is mounted directly to the other end of the shaft. The frame has been developed from common components such as square hollow section and aluminium plates to keep cost down, as well as see through polycarbonate to improve the aesthetics.

The key performance metrics are the size restrictions, particle size and speed of operation. Firstly the size of the shredder is 292.9 x 194.6 x 346.0mm in depth, width and height respectively, when the motor set up is included this increases to 510.6mm depth but this is not considered in the space requirements. To ensure that all particles are no larger than 5mm in any direction the shredder includes a grate with 4mm diameter holes. The grate follows the curvature of the blades and prevents any large plastic from entering the collection tub. The speed of operation can be approximated assuming 60rpm each 375mL bottle would need to be reduced in 10 full rotations, which is reasonable as the blade spacing is an aggressive 3mm.

To analyse the shredder all critical components have been studied and designed with appropriate safety factors. Some critical components underwent an additional FEA simulation on Solidworks and Ansys, this includes the blades and shaft which was further optimised by hollowing out the material. The dominant materials used for the shredder are mild steel and aluminium, as they provide a good balance of cost, weight and strength as required for this application. They are also easy to work with and can be machined or manufactured easily.

The cost per shredder is estimated to be \$497.99, which falls just within the \$500 price range, the estimated cost of 50 units is \$24899.50. The cost of the housing and the cutting elements is just over \$200 each while the handle and gear mechanism cost almost \$70. These include purchased and fabricated components, with materials and manufacturing costs included.

Additional design considerations include safety, such as the lid and funnel to ensure that users are not endangered by the machine. The design also takes sustainability into account, being developed from metals which can be easily recycled, the design is also modular and can be taken apart to replace faulty components and increase the lifespan of the whole machine.

The final shredder design meets all design specifications and makes for an effective, safe and user friendly household plastic shredding machine.

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2 Introduction

Plastic waste is a major issue in today's society, contributing to significant pollution and damage to our ecosystem. The incineration, burying, and littering of otherwise recyclable plastic not only damages the environment directly, but promotes further plastic production, perpetuating a negative cycle. Our team was tasked with designing a household plastic shredder or grinder. The grinder is meant to be a small, easy to use, low barrier to entry appliance in the home in order to promote plastic recycling at the individual level. The plastic produced is to be converted to 3D printing filament for the production of common household items, ultimately reducing plastic production. This report explores the design of the shredder, justification for design decisions, and analysis conducted.

The shredder is required to have the capacity to process up to six 375mL plastic drinking bottles per minute, reducing them to pieces no larger than 5mm in any direction. It must fit within a L500 W500 H400 mm space, and cost no more than \$500 each (excluding the motor) when producing a small run of 50 prototypes. The user can manually operate the shredder using their own power, but an electric mode must also be available for convenience.

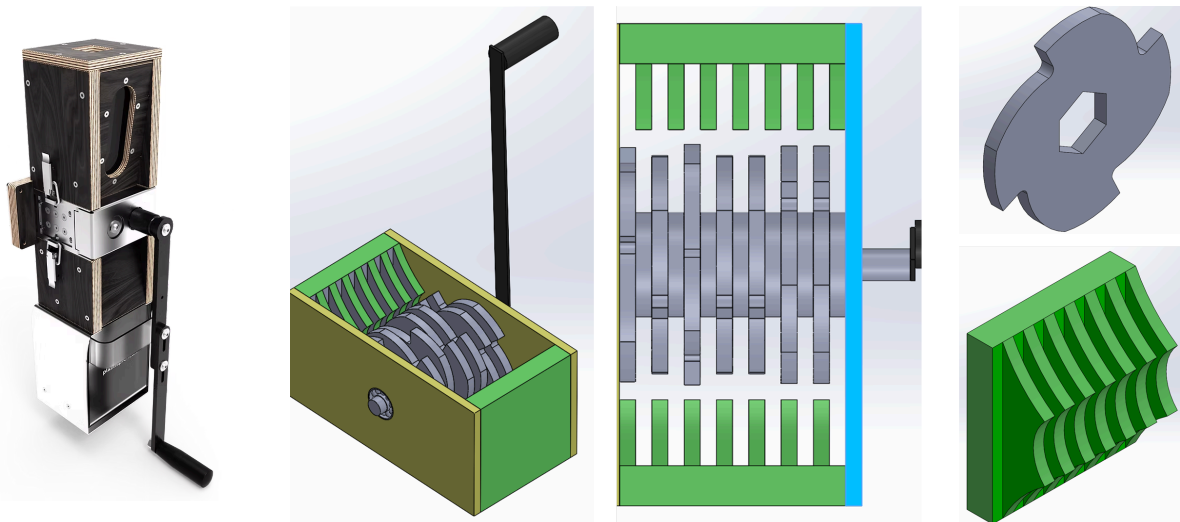


Figure 2.1: Examples of shredder designs to help provide an understanding of the product

Throughout the design process, there were design considerations and approaches to improve the end user experience. The system was designed with safety, serviceability, and longevity in mind. Ergonomics had to be considered to the extent of ease of transport, as well as the method of power input.

The team's approach to the project involved weekly meetings to discuss design changes, project direction, and progress towards internal deadlines. These meetings doubled as design reviews, in which our engineers presented their components and their design justifications for review and feedback from the broader team. This process led to a cohesive system with strict adherence to the requirements.

3 Design Progression

3.1 Initial Concept and Preliminary CAD

Our preliminary design for the plastic shredder used a single shaft bladed mechanism that meshed with the teeth for a spacing of 3mm. The design also included a curved grate with 4mm holes so that all plastic would be within the size tolerance and larger chunks to be reground. The images below show the completed assembly and the exploded view for the cutting mechanism, and the housing.

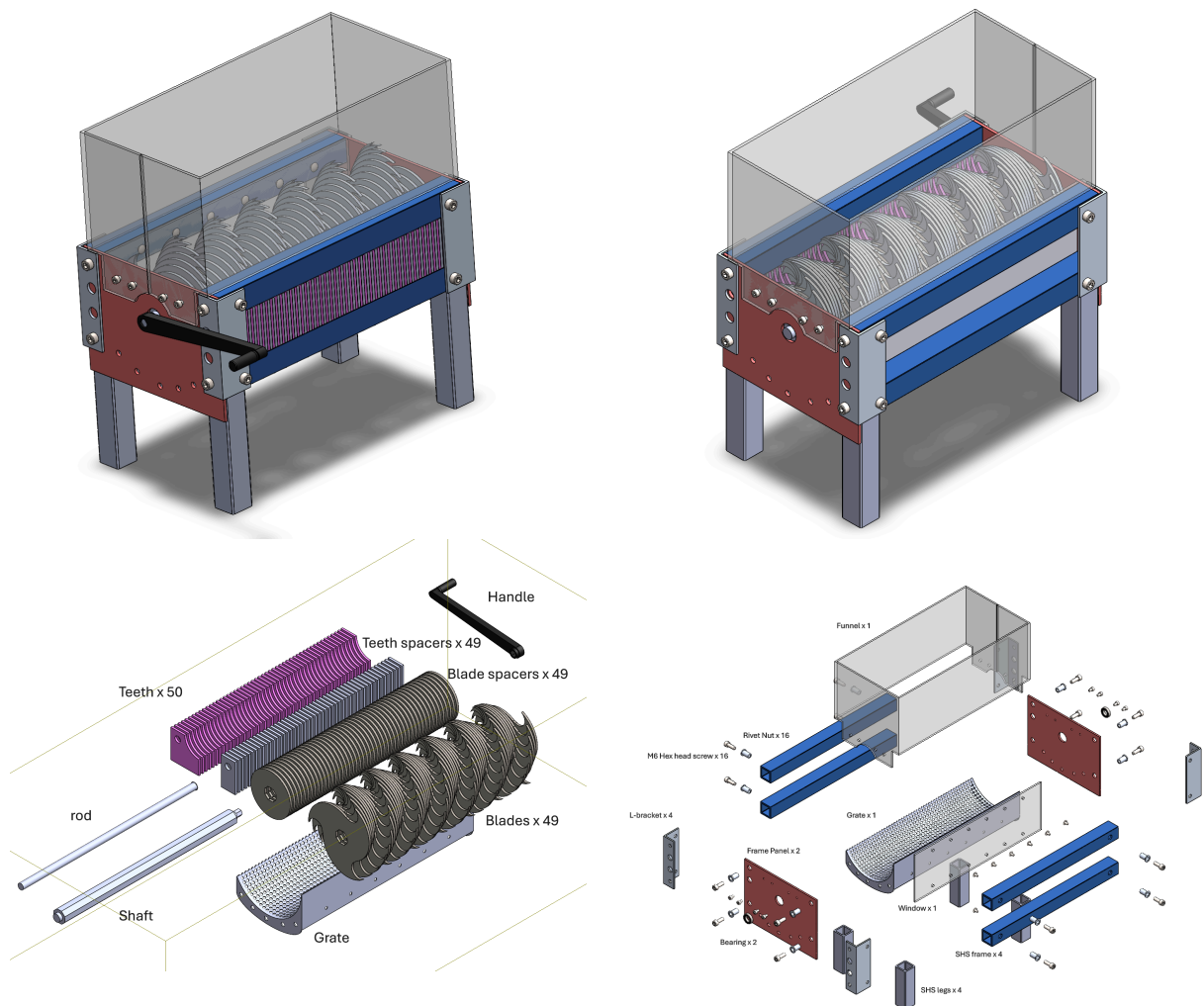


Figure 3.1.1: Multiple views of the CAD from the Preliminary Design

3.2 Changes from the Preliminary Design

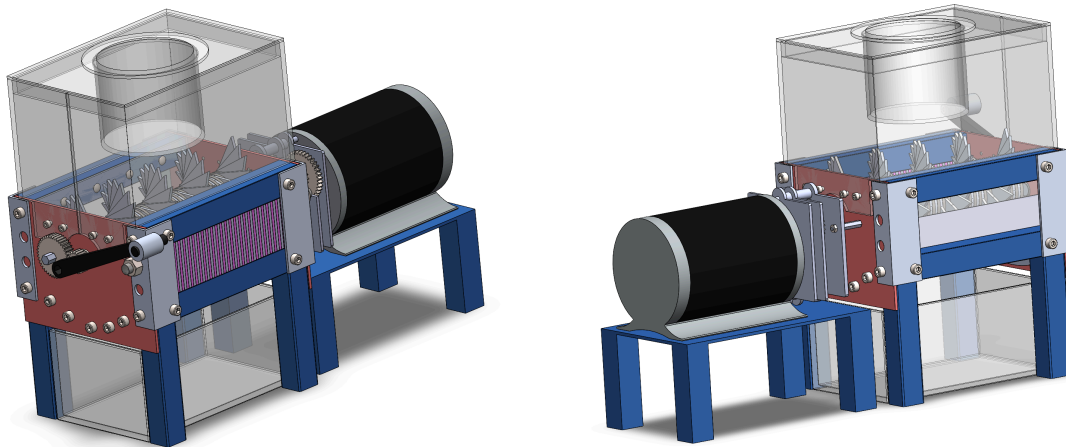


Figure 3.2.1: CAD of the Final Design after changes

The preliminary design was an essential baseline for the final shredder created by Group 305. Upon FEA analysis, alongside general calculations for gears and bearings, some preliminary components could be optimised to increase operational efficiency and lifespan.

3.2.1 - Alterations of Components that Appeared in Preliminary Submission

Changing Shredder Length - One of the first changes of the design that took place after the preliminary design was made was altering the length of the shredder. The shredder as a whole was shortened so that at any given time, only a maximum of 4 blades will be shredding plastic at a time, which drastically reduced the maximum torque that the shredder could experience at any given time. This change reduced the amount of blades on the shaft from being 49 to an even 32, meaning at each rotation of the blades on the shaft, there are now the same amount of blades. These changes reduced the internal length of the shredder from 307mm to 201.6mm, essentially reducing the length of the shaft by a third. Alongside reducing the amount of space the shredder now takes up in a household, the new length of the shredder is closer to the size required to shred 375mL plastic bottles, and reduces the unutilised space in the design.

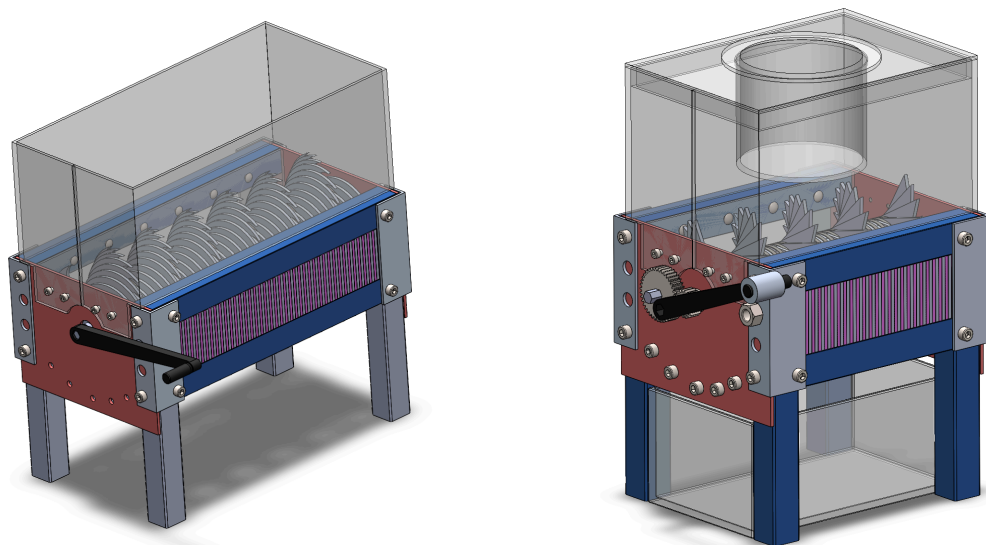


Figure 3.2.1.1: CAD views highlighting the difference in length between the Preliminary (left) and Final (right) designs.

Blade Geometry - The blades' original round shape at the sharp ends was changed to a blade that was much flatter at the ends, which reduced the overall sharpness of the blades, but also allowed for a larger surface area of the blade to come in contact with the plastic bottle, improving the cutting efficiency. The exact dimensions of the blade were determined using FEA analysis and Solidworks' Optimization feature (See 4.9 for more details).

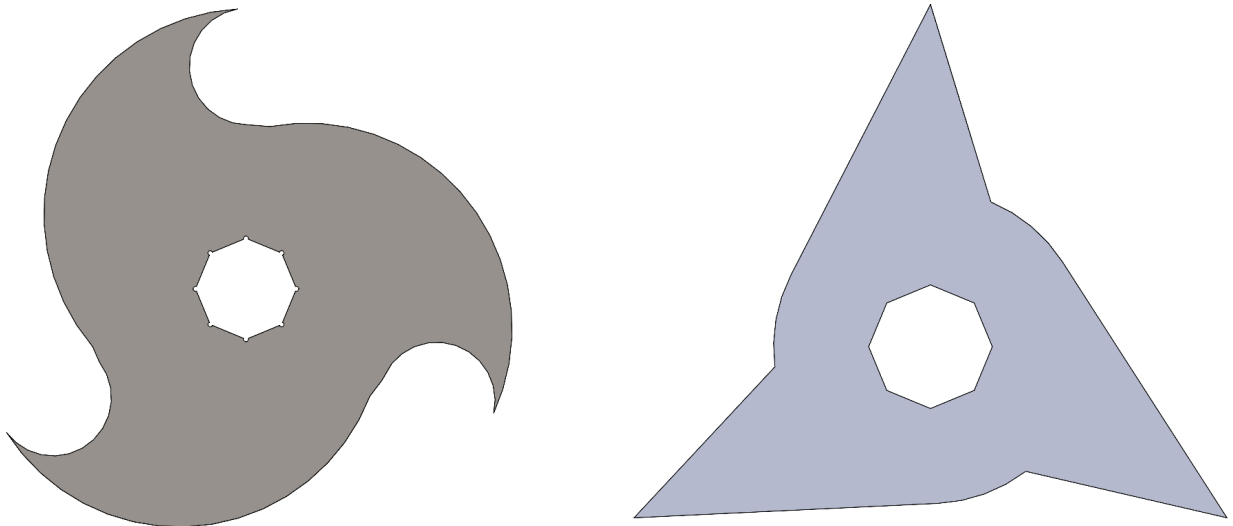


Figure 3.2.1.2: Comparison the the blade geometry from the Preliminary (left) and Final (right) designs

Shaft Redesign - Rather than creating the shaft as a singular piece, a decision was made to hollow out the inside of the shaft, and make the ends of the shafts screw on to the shaft with the use of threaded fasteners. This design decision was made to reduce the weight of the shaft, whilst also allowing for greater maintenance of smaller components of the shredder and allow for the design to be manufactured easily. Additionally, the threaded fasteners on each side of the shaft differ in the direction that they screw, as to prevent loosening the assembly during operation whilst allowing the blades to always rotate the correct direction.

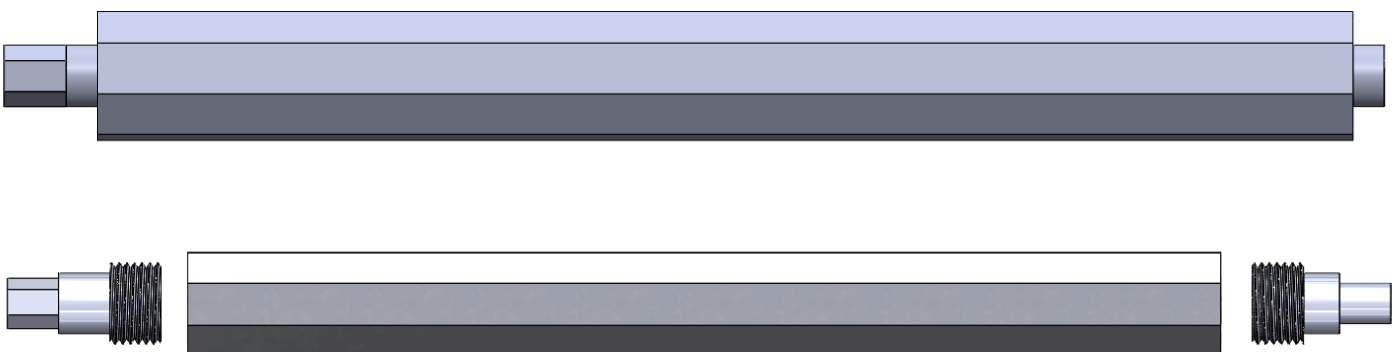


Figure 3.2.1.3: CAD showing how the shaft was changed to multiple components in the final design.

Handle Redesign - The length of the handle was slightly adjusted so it would not interact with the ground. The grip was also modified to rotate freely as well ,achieved by press fitting the grip onto a bearing, and assembling the handle from multiple parts. These changes directly improve the ergonomics and user experience. Additionally, the redesign of the handle streamlined manufacturing by reducing the complexity of the part.

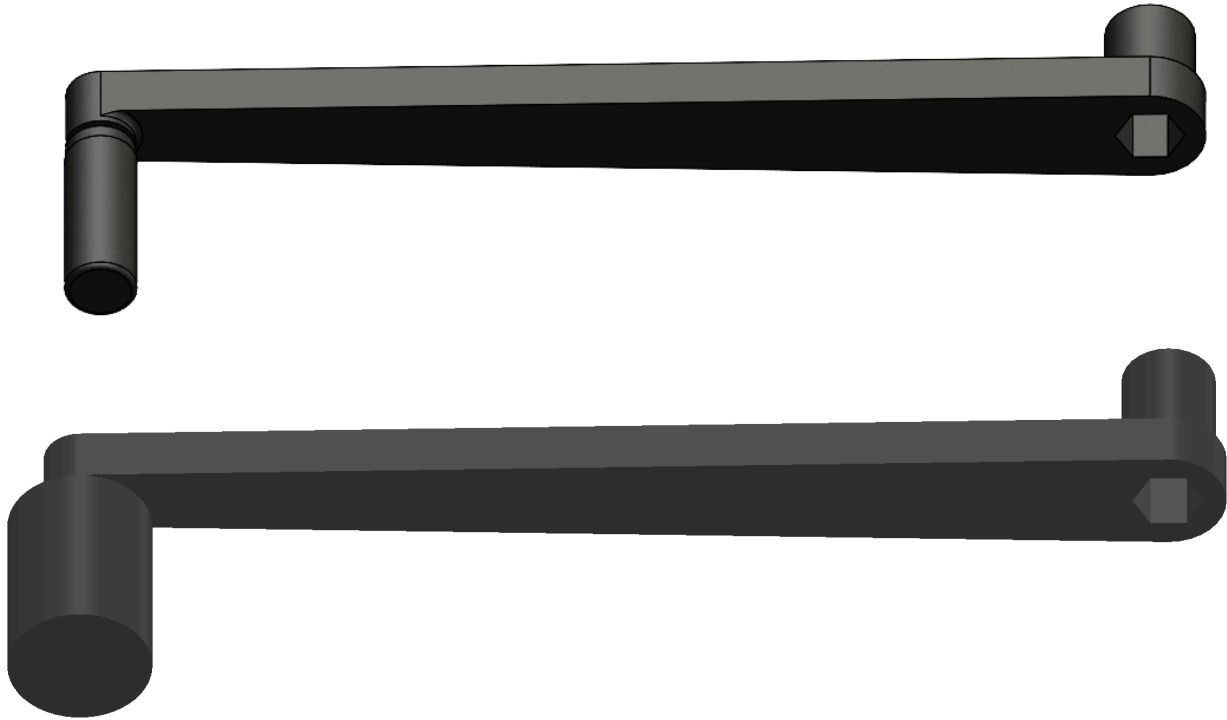


Figure 3.2.1.4: Comparison of the handle from the Preliminary (top) and Final (bottom) designs

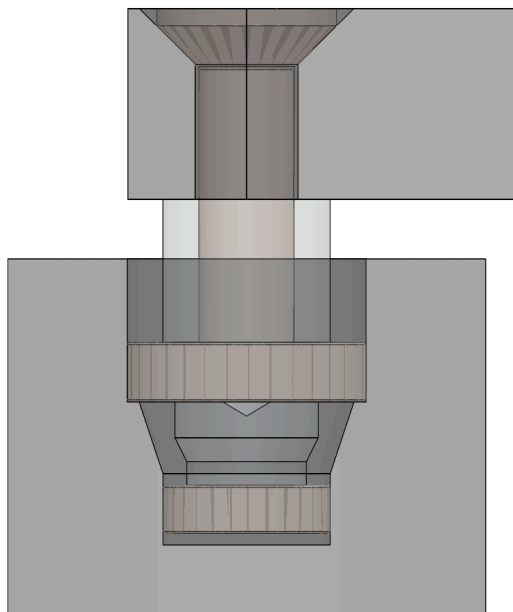


Figure 3.2.1.5: CAD of the assembly of the handle grip

3.2.2 - Addition of Components for the Final Design

Addition of a Lid - The preliminary design included a Polycarbonate funnel above the blades to put space in between the operator's hands and any potential danger, but introducing a lid furthered this safety measure by making it harder for someone's hands to touch the blades unintentionally. The lid above these walls is also made of polycarbonate, and includes a hole in the centre that a topper fills while the shredder is not in operation. When the shredder is running, the topper can be taken out of the lid, allowing for a plastic bottle to enter the shredder, with the topper then being used to push the plastic bottle further into the shredder and speed up the shredding process, similar to what you would see in a food processor.

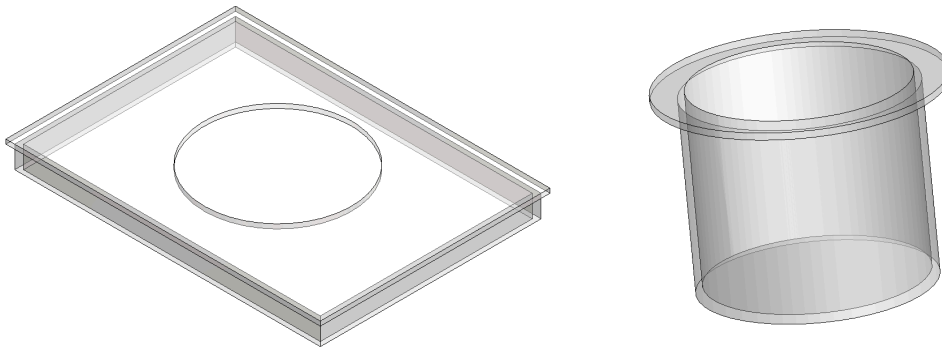


Figure 3.2.2.1: CAD of the polycarbonate lid and topper

Addition of a Container - Something that was missing from the preliminary design was a way to store any of the plastic pieces that were capable of falling through the grate and hence meeting the size requirements to be recycled for 3D printing use. The container that has been designed takes up as much surface area at the bottom of the shredder as it possibly can, preventing any plastic from falling through the grate and then falling outside of the container. While the container takes up a large amount of the bottom surface area of the shredder, it has also been designed in a way that it can still very easily slide out from the bottom of the shredder so that small plastic pieces can be easily extracted after the shredding process has been completed.

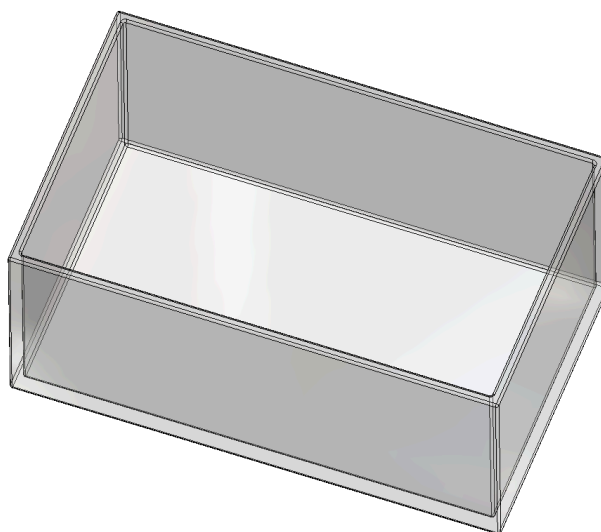


Figure 3.2.2.2: CAD of the polycarbonate collection tub

Gear Ratios on Manual Operating Side - Whilst completing load analysis for the preliminary submission, it became clear that some people may struggle to produce enough force to fully operate the shredder, even after the maximum torque decreased as the design was shortened. To fix this, a small gear design was attached onto the manual side of the shredder, giving users the choice of two different gears to attach the handle to and run the device.

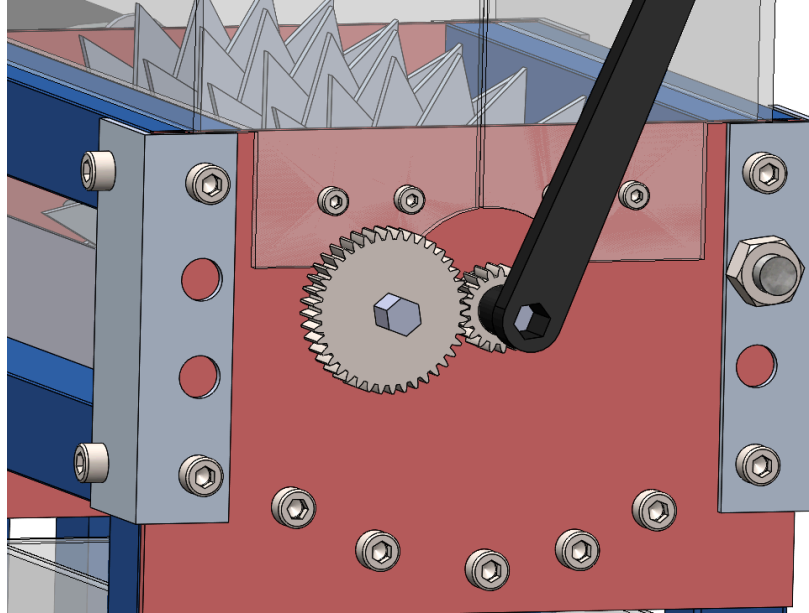


Figure 3.2.2.3: CAD of the handle-side gears, providing the user optional additional torque

Addition of a Motor - A reasonable motor for the size of the shredder was selected to allow for the shredder to be capable of operating autonomously. To meet the required torque of the shredder, a gear box was designed to be placed in between the motor and the shaft to increase the output torque that would be transmitted to the shaft. This is discussed further in section 4.2.

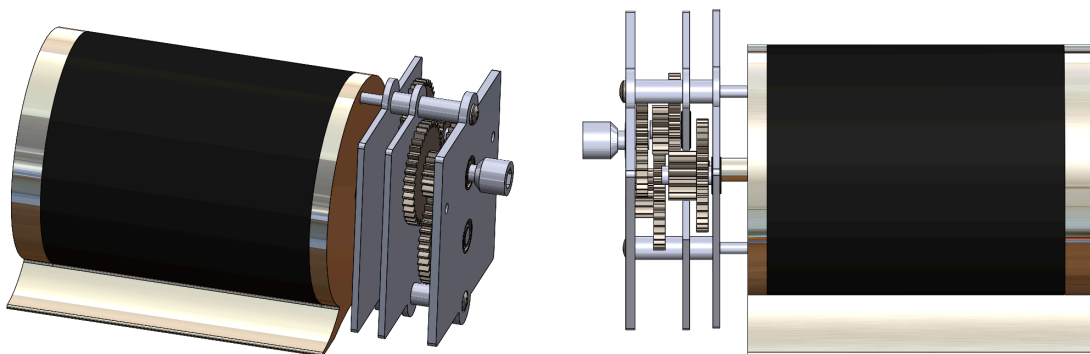
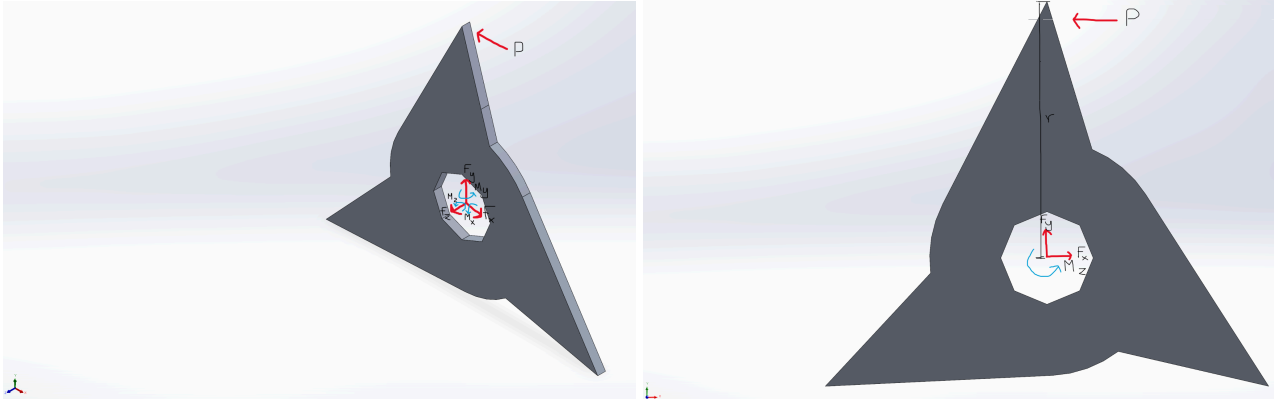


Figure 3.2.2.4: CAD of the bespoke gear box and commercially sourced electric motor.

4 Design Analysis and Component Selection

4.1 Load Analysis

4.1.1 Blade



$P=162\text{N}$: assumed worst case scenario force $r=60\text{mm}=0.06\text{m}$

2 dimension problem with:

$$F_z = F_y = M_y = M_x = 0$$

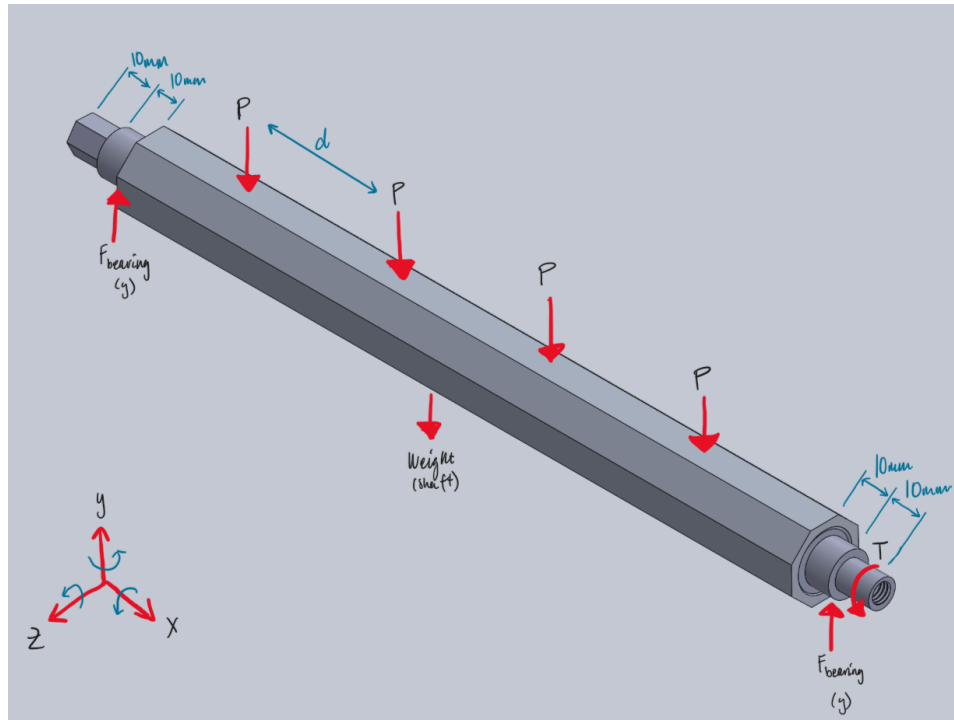
$$\Sigma F_x = 0 = F_x - P \qquad \Sigma M_z = 0 = M_z + P \times r$$

$$P = 162\text{N} \qquad P \times r = 162 \times 0.06$$

$$F_x = 162\text{N} \qquad M_z = -9.72\text{Nm}$$

Comprehensive stress analysis for the blades will be presented in the FEA section of the report.

4.1.2 Shaft



$$P = 162N \quad W_{shaft} = 0.11789 \times 9.81 = 1.15N \quad d = 0.04032m$$

2 dimension problem with:

$$F_z = F_x = M_y = M_z = 0$$

$$\Sigma F_y = 0 = 4 \times P + W_{shaft} - 2 \times F_{bearing(y)}$$

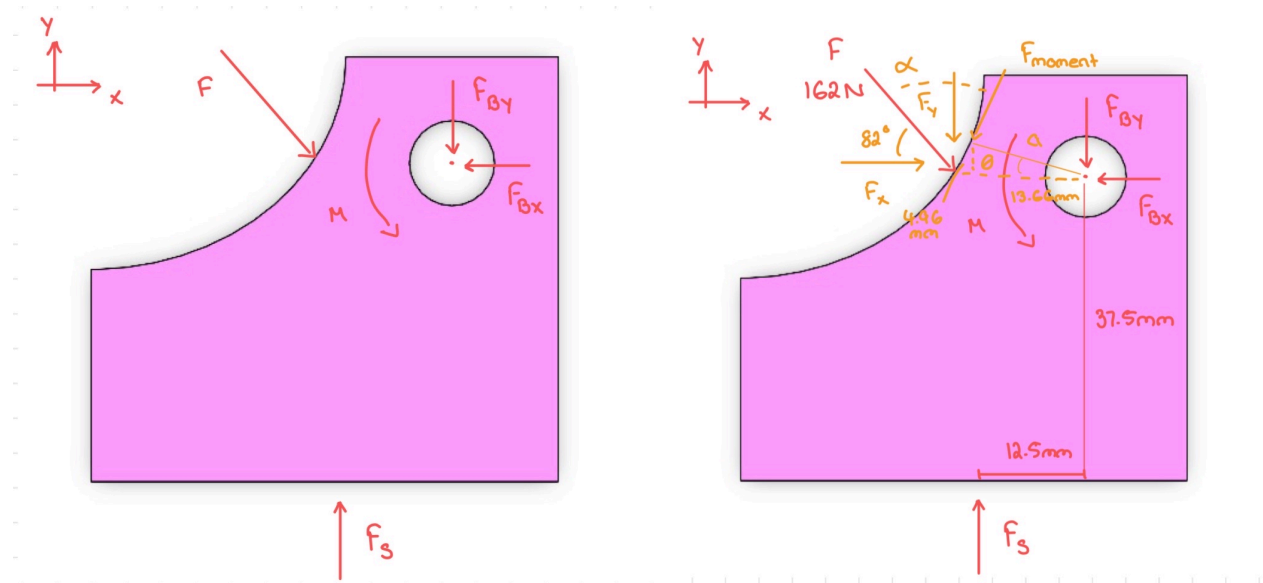
$$F_{bearing(y)} = 324.58N$$

$$M_{max} = \frac{w \times l^2}{8} \approx \frac{4P}{d} \times \frac{d^2}{8}$$

$$M_z = -3.27Nm$$

Comprehensive stress analysis for the shaft will be presented in the FEA section of the report.

4.1.3 Teeth



a and θ calculation

$$a = \sqrt{4.96^2 + 13.66^2}$$

$$= 14.63 \text{ mm}$$

$$\theta = \tan^{-1}\left(\frac{4.96}{13.66}\right)$$

$$= 19.96^\circ$$

α and F_{moment}

$$\alpha = (90 - 82) + 19.96$$

$$= 29.96^\circ$$

$$F_{\text{moment}} = \cos(29.96^\circ) \times 162$$

$$= 143.1 \text{ N}$$

$\sum F_x = 0$

$$F_{Bx} = F_x = 162 \times \cos(82^\circ)$$

$$= 22.5 \text{ N}$$

$$* F_x = 22.5 \text{ N}$$

$$F_y = 160.4 \text{ N}$$

$\sum M = 0$ and $\sum F_y = 0$

$$\sum M = a \times F_{\text{moment}} - 12.5 \times F_s = 0$$

$$F_s = \frac{14.53 \times 143}{12.5}$$

$$= 166.3 \text{ N}$$

$$\sum F_y = -F_y - F_{By} + F_s = 0$$

$$F_{By} = -F_y + F_s = -162 \times \sin(82^\circ) + 166.3$$

$$= 5.88 \text{ N}$$

F_{ROD}

$$F_{\text{ROD}} = \sqrt{F_{Bx}^2 + F_{By}^2}$$

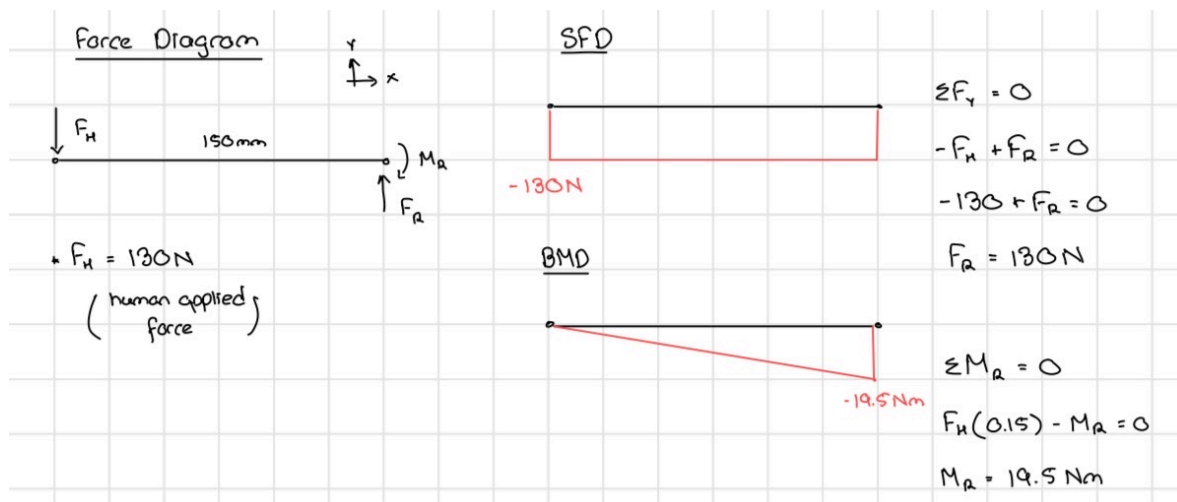
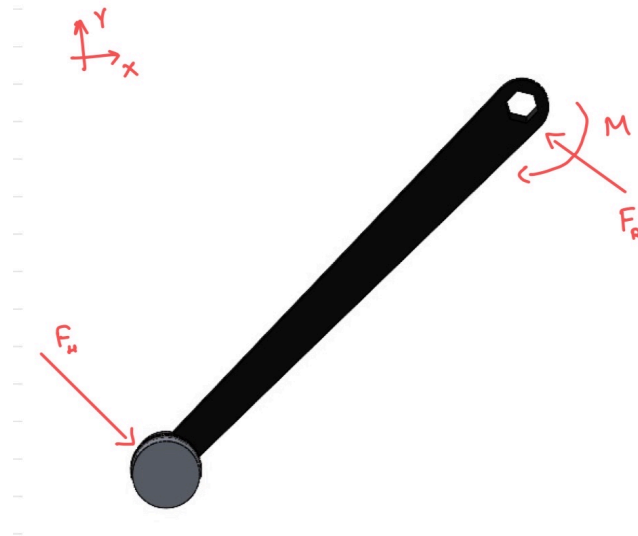
$$= \sqrt{22.5^2 + 5.88^2}$$

$$= 23.3 \text{ N}$$

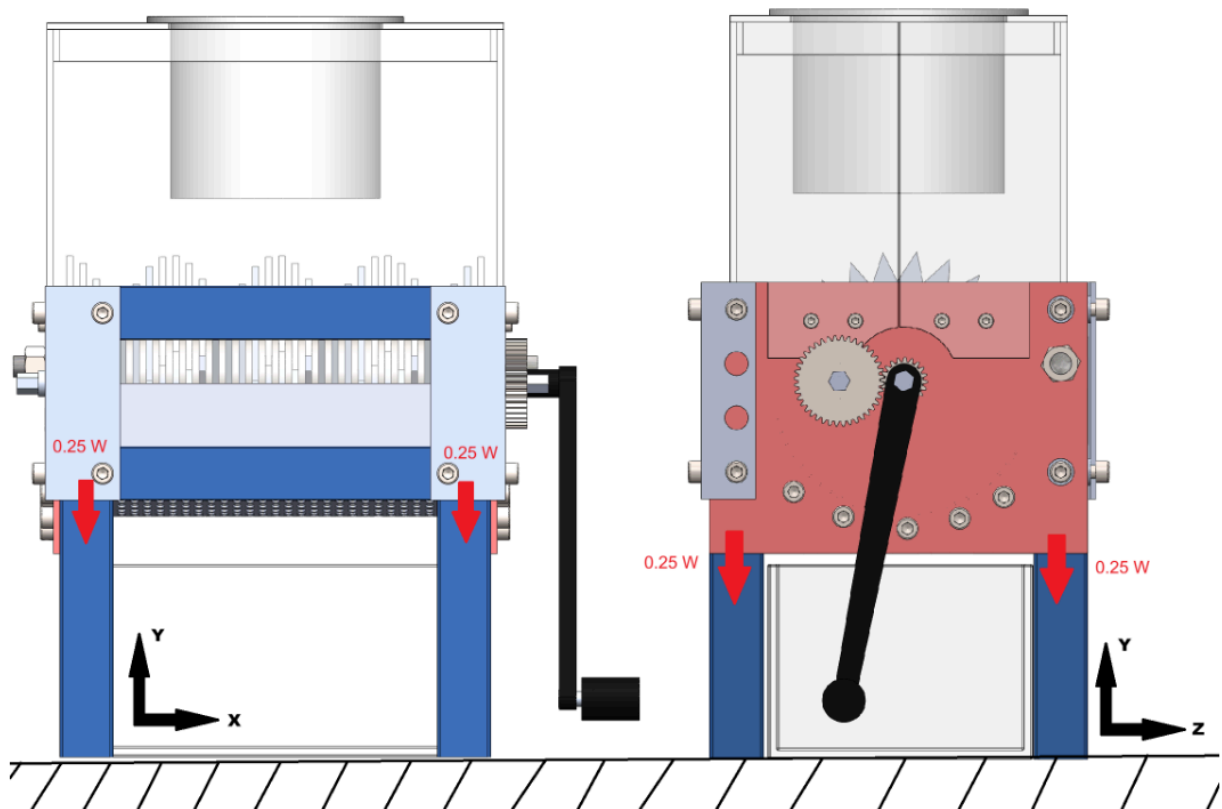
$$\beta = \tan^{-1}\left(\frac{F_{By}}{F_{Bx}}\right)$$

$$= 14.65^\circ$$

4.1.4 Handle



4.1.5 Legs



The supported weight of the grinder, W , is 4.6 kg

With each leg supporting a quarter of the weight, that is:

$$0.25W = 0.25 \times 4.6\text{kg} = 1.15\text{kg}$$

or

$$1.15\text{kg} \times 9.81\text{m/s} = 11.28\text{N}$$

On each leg, $F_y = 11.28\text{N}$

4.2 Motor and Gear Selection

The motor selected was the Dayton MY1020, a 1000W, 3000rpm electric motor. It was chosen due to its reasonable torque output, small size factor, and relatively low cost (see Appendix A). The torque output was calculated by:

$$T = 60 \times \frac{\text{Power}}{2\pi \times \text{rpm}} = 60 \times \frac{1000}{2\pi \times 3000} = 3.18 \text{ Nm}$$

The gearbox was designed to adapt the 3000 RPM, 3.2 Nm motor for the high torque requirement of a plastic shredder using a 4 stage compound gear train. Each stage was made of identical 15:37 spur gear pairs which combined to create a total gear ratio of 37.02:1 $(37/15)^4$. This reduction slows the output speed to 81 RPM $(3000/37.02)$ and increases the torque to 118.4 $(3.2 \text{ Nm} \times 37.02)$, which achieves the required Factor of Safety of 3.0 against the shredder's operational need of 39.4 Nm.

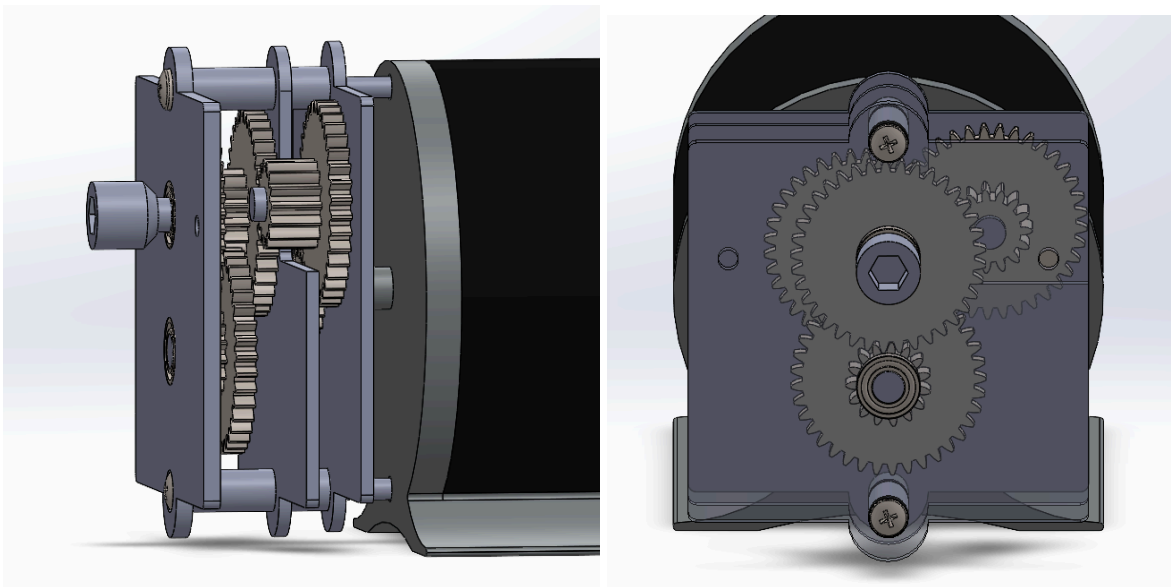


Figure 4.2.1: CAD of the gearbox assembly onto the face of the electric motor.

The gearbox was designed to be compact and mounted directly onto the motor and plastic shredder. This allowed for precise alignment between each input and output shaft. If misaligned, the shafts would spin off centre at an angle, creating friction and vibrations. This misalignment can cause uneven load distribution on components, leading to premature failure. For example, bearings may overheat and be damaged or shafts may fatigue and break. The increased friction can also waste energy, reducing power efficiency and create loud unwanted noises.

The gearbox allows for the use of a smaller, faster and cheaper motor. The cost of electric motors are mainly driven by its torque output which is due to its size, larger magnets and thicker copper windings. A gearbox solves this by acting as a torque multiplier, converting cheap, high speed and low torque inputs from a smaller motor to a lower speed and high torque output.

The handle has a gear attachment intended to provide more advantage for users who struggle with force output. The selection has a 40 tooth input gear, connected to a 20 tooth gear on the shaft. Both gears have hex key attachments so the handle can be easily moved between the gears by the user. The 20 tooth gear mounts directly to the shaft and is therefore 1 to 1, whereas the 40 tooth gear allows 2 handle rotations to 1 shaft rotation.

The gears are heat treated and have the values below, with a force of 130N from the handle.

$$\sigma_Y = 490\text{MPa} \quad Y_{20} = 0.315 \quad Y_{40} = 0.385 \quad m = 1\text{mm} \quad b = 10\text{mm}$$

$$T_{\text{handle}} = 130 \times 0.15 = 19.5\text{Nm}$$

The force, stress and safety factor of the 40 tooth gear.

$$F_{40} = \frac{T}{r} = \frac{19.5}{0.02} = 975\text{N} \quad \sigma_{40} = \frac{F_{40}}{bmY_{40}} = \frac{975}{10 \times 1 \times 0.385} = 253.2\text{MPa}$$

$$SF = \frac{\sigma_Y}{\sigma_{40}} = \frac{490}{253.2} = 1.94$$

The force, stress and safety factor of the 20 tooth gear.

$$F_{20} = F_{40} = 975\text{N} \quad \sigma_{20} = \frac{F_{20}}{bmY_{20}} = \frac{975}{10 \times 1 \times 0.315} = 309.5\text{MPa} \quad SF = \frac{\sigma_Y}{\sigma_{40}} = \frac{490}{309.5} = 1.58$$

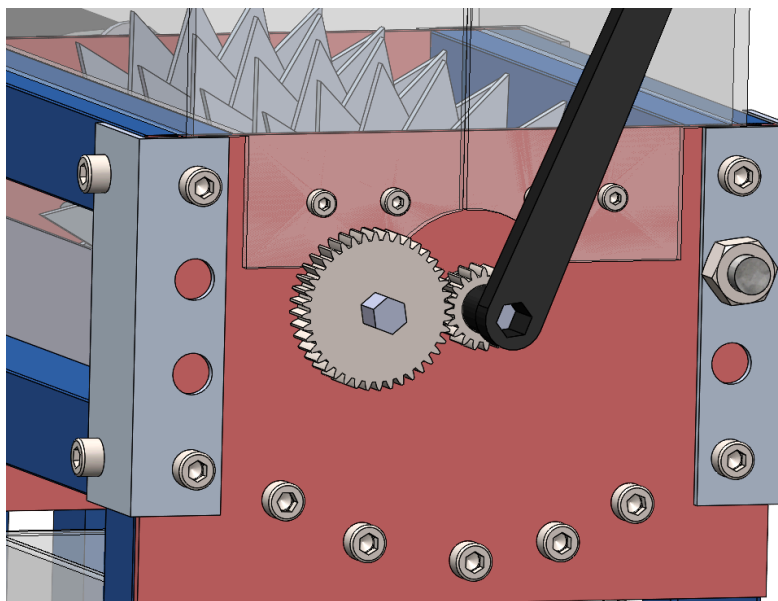


Figure 4.2.2: Handle-side gears provide optional extra torque.

While the safety factors of the gears are low, they are only intended to be used by users who require mechanical advantage, therefore the use will be low and the forces in line with the assumed 130N. The extra mechanical advantage would also allow higher torque on the shaft and allow the user to break through larger pieces of plastic or to un-jam the machine. The gears are mounted on threaded shafts, so to replace a broken gear the shaft would be unthreaded and then the gear could easily slide off the shaft and be replaced with a new gear, maintenance is a key consideration for the gear design.

4.3 Shaft Design and Analysis

Shaft Design and Analysis

The main shaft was designed with an assumed diameter of 20 mm (inner diameter of the octagonal cross-section).

Several design and manufacturing factors were considered the geometry of the shaft. The manufacturing process strongly influenced design constraints due to the unique end geometries and hollow internal structure, both of which were selected to reduce overall weight and material costs while maintaining adequate torsional rigidity. The blade mounting system utilised an octagonal interface, allowing each blade to fit securely onto the shaft in a jigsaw-like configuration where each blade is rotated by 135°. Although this design introduced localised stress concentrations, finite element analysis (FEA) confirmed that the resulting stresses remained within acceptable limits. To improve performance and durability, the shaft length was minimised to optimise torque transfer, reduce fatigue, and limit deflection, while simultaneously lowering production costs. The shaft design also accommodates functional integration site on each end, allowing the handle and motor to be mounted on opposite sides. Reliable torque transmission between components was achieved using press/shrink bearings on the shaft that locate within the housing unit. Finally, the design ensured that the centre of mass coincided with the centre of rotation, promoting balanced and smooth operation during use.

4.3.1 Shaft Sizing and Stress Analysis

The maximum combined bending and torsional stress was estimated using the modified ASME code for shaft design. The maximum shear stress is given by:

$$\sigma_{\max} = \frac{16}{\pi d^3} \sqrt{M^2 + T^2}$$

$$\sigma_{\max} = 75.48 \text{ kN/m}^2$$

The shaft was designed with a factor of safety (FoS) of 1.27 to ensure reliability under expected operating loads. Further explored in FEA

The endurance limit was determined as:

$$S_n = (0.5 \times 570 \text{ MPa})(1)(0.9)(0.8)(1)(0.868) = 176.47 \text{ Mpa}$$

Assuming a machined surface finish factor of 0.8 and a 95% reliability factor of 0.868 (standard for rotating machinery), this value was used in the fatigue evaluation.

Using the ASME code relationship:

$$d = \left[\frac{32 \times \text{FoS}}{\pi} \sqrt{\left(\frac{M}{S_y}\right)^2 + \frac{3}{4} \left(\frac{T}{S_y}\right)^2} \right]^{1/3}$$

$$d = 0.01382 \text{ m}$$

The recommended shaft diameter was 13.8 mm, which falls close to the assumed 20 mm diameter. Following additional FEA validation, the 20 mm dimension was retained to improve stiffness and reduce deflection.

4.3.2 Deflection and Critical Speed

The shaft's moment of inertia was calculated as:

$$I = \frac{\pi d^4}{64} = 7.854 \times 10^{-9} \text{ m}^4$$

With a modulus of elasticity $E = 7.2 \times 10^{10} \text{ N/m}^2$, and assuming a uniformly distributed load from the blades, the midspan deflection was determined using:

$$\delta = \frac{PL^3}{48EI} = 1.96 \times 10^{-4} \text{ m} = 0.196 \text{ mm}$$

The critical speed was then evaluated using:

$$\omega_c = \sqrt{\frac{g}{\delta}} = 223.74 \text{ rad/s}$$

Thus, the optimal operating speed is approximately 111.87 rad/s ($\approx 1068 \text{ RPM}$). In operation, shredding six bottles would require about 60 rotations, roughly 1 rotation per second, keeping the speed well below the critical limit when hand-cranked. The motor can also be restricted to not exceed this limit to avoid resonance and vibration issues.

4.3.3 Fatigue and Failure Considerations

FEA results indicated that fatigue stresses along the shaft remain low, with an expected life of approximately 4×10^7 cycles before fatigue failure becomes critical. The hollow configuration reduces cyclic stresses while improving weight efficiency.

4.3.4 Maintenance and Accessibility

Maintenance procedures were designed for ease and cost-effectiveness. The connection points can be removed when signs of wear or fatigue appear, and blades and spacers can be slid off individually without replacing the entire shaft assembly. This modular approach simplifies servicing and reduces material waste. A summary of the maintenance procedure is provided in **Section 5.3**.

4.3.5 Shaft Tolerances and Bearing Fit

To ensure proper bearing alignment, the shaft diameter and tolerance were designed to match the bearing bore size. The bearings used are 61701-2RS, with specifications:

- Bore diameter: 12 mm
- Outer diameter: 18 mm
- Width: 4 mm

Assuming an inner bearing diameter of 12 mm, the corresponding shaft journal was designed to achieve a transition or light interference fit, as discussed in **Section 4.8.1** (Connection of the Shaft to the Bearing Inner Ring).

4.3.6 Cost, Environmental, and Material Considerations

Material selection prioritised cost efficiency, recyclability, and low environmental impact. The hollow design reduced material usage and machining time, lowering both production cost and energy consumption. Manufacturing the shaft from a readily available steel alloy ensured durability, machinability, and sustainability through the use of recyclable materials.

4.4 Bearing Selection and Analysis

Bearings serve an important part of the design and allow for the regulated rotation of parts in the shredder assembly, hence there are a number throughout our design. All the bearings are rolling element ball bearings as they allow for easier connection to the design than journal bearings, more importantly they will be operating at low speeds (approximately 60rpm) which is ideal for rolling element bearings due to uneven distribution of fluid in the journal bearing at low speed and during the transient state. They are also all radial ball bearings since the applied force in all cases is acting perpendicular to the central axis, however all are deep groove ball bearings which do allow for small axial loads that might take place in uneven operating conditions. Although ball bearings are louder than journal bearings they will be operating at low speeds and the noise of the crushing plastic will mask it.

4.4.1 Handle grip: DDL-1060ZZ

Bore = 6mm Outer diameter = 10mm Width = 3mm

This was chosen as a small standard bearing was required and experiences little force so a cheap and simple option was chosen. The load ratings have been taken from the item specifications on RS online.

$F_r = 130\text{N}$ (human applied force)

$K_r = 1$ (90% reliability factor)

$K_a = 1.5$ (light impact ball bearing assume higher end for worst case)

$L_r = 90 \times 10^6$

$C = 457\text{N}$

$C_{\text{static}} = 194\text{kN}$

$$L = K_r L_r \left(\frac{C}{F_r K_a} \right)^{3.33}$$
$$L = 1 \times 90 \times 10^6 \times \left(\frac{457}{130 \times 1.5} \right)^{3.33}$$
$$L = 1.49 \times 10^9 \text{ cycles}$$
$$FOS = \frac{C_{\text{static}}}{F_r}$$
$$FOS = \frac{194000}{130}$$
$$FOS = 1492$$

4.4.2 Shaft and Gear: 61701-2RS

Bore = 12mm Outer diameter = 18mm Width = 4mm

The ends of the shaft only had a very thin space for the bearing to fit between the spinning components inside the shredder and the location where the handle attaches so a thin bearing had to be selected. This slim design was also applied to the gears as they suffered the same requirements so the same bearing was selected. This means 3 are required per device, one on each end of the shaft and one more for the gears. They are also enclosed (the balls are behind rubber) which should prevent any small shards of plastic from the shredder from jamming the bearing since these will have a face in the internal compartment of the shredder. The load ratings have been taken from an information page on C&H bearings.

$F_r = 417\text{N}$ (maximum reaction force at ends of shaft (see FBD analysis))

$K_r = 1$ (90% reliability factor)

$K_a = 1$ (gearing and uniform load)

$$L_r = 90 \times 10^6$$

$$C = 926 \text{ N}$$

$$C_{\text{static}} = 530 \text{ N}$$

$$L = K_r L_r \left(\frac{C}{F_r K_a} \right)^{3.33}$$

$$L = 1 \times 90 \times 10^6 \times \left(\frac{926}{417 \times 1} \right)^{3.33}$$

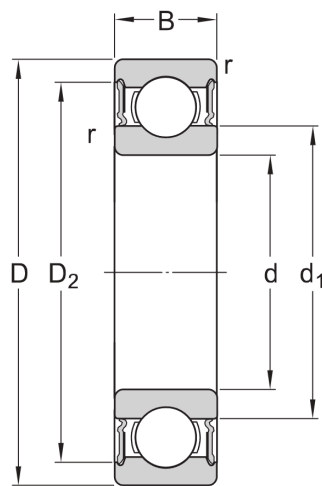
$$L = 1.3 \times 10^9 \text{ cycles}$$

$$FOS = \frac{C_{\text{static}}}{F_r}$$

$$FOS = \frac{530}{417}$$

$$FOS = 1.27$$

Although the FOS of 1.27 seems quite low it is a very conservative approach assuming the absolute worst case scenario in the shredder system typically in use the force would be much lower so it should not cause a problem.



Dimensions

d	12 mm	Bore diameter
D	18 mm	Outside diameter
B	4 mm	Width
d ₁	13.9 mm	Shoulder diameter inner ring
D ₂	16.7 mm	Recess diameter outer ring
r	0.2 mm	Chamfer dimension

Performance

C _r	0.926 kN	Basic dynamic load rating
C _{0r}	0.53 kN	Basic static load rating
	13000 r/min	Limiting speed

Weight

≈m	0.0031 kg	Weight
----	-----------	--------

All of these bearings can endure a very large amount of cycles under the current operating conditions. Assuming rpm=60 and runs for a minute a day (5 bottles - above average household use of 4) the lifespan will well exceed requirement, lasting thousands of years before failing so fatigue and failure is not a concern. If something were to happen and a bearing got a piece of plastic jammed in it the bearings are accessible from the exterior so relatively easy to access for maintenance or replacement.

4.4.4 Further research - Magnetic bearings

A more advanced option is using magnetic bearings, a device heavily researched and produced by SKF (Swedish bearing Factory). This magic of this technology is that there is no contact between the bearing and the shaft due to the magnets levitation it in place, which eliminates friction and wear and tear essentially requiring no maintenance, sensors in the bearing also allow for real time active damping of the shaft to minimise vibrations which further reduces wear and tear on other components on the shaft perfect for high speed high force operation¹. However the benefits are mismatched with the needs for our project; wear and tear isn't a concern of ours due to the small loads and operating speeds and more importantly magnetic bearings require a constant current to run through the coils inducing the magnetic field which would add unnecessary cost and complexity to our design, hence we opted for more simpler ball bearings.

1 - Bearings for Aerospace Applications (Rejith. R, Kesavan, D., Chakravarthy. P, & S.V.S. Narayana Murty)

4.5 Welding and Fastener Selection

4.5.1 SHS Leg welds

The square hollow sections are made from aluminium so a TIG weld using E4043 aluminium filler as required to join aluminum sections. The welds are on both sides of the leg and join the horizontal square hollow section to the legs. These experience parallel loading at a peak of 130N when the user is pushing on the handle in the horizontal position assuming the legs are fixed to the floor. They are very short welds hence TIG welding has been selected due to its precision and these welds will hold comfortably since the FOS is very high, however because the welds are so small optimising them would not yield much cost benefit so the FOS of 41 is acceptable.

$$\text{Length} = L = 22\text{mm} \quad S_{yt} = 20\text{kpsi} \quad S_{sy} = 0.58 \times 20 = 11.5\text{kpsi} = 80\text{MPa} \quad F=130\text{N}$$

$$\text{let } h = \text{SHS thickness} = 3\text{mm} \quad t = 0.707 \cdot h = 2.1\text{mm} \quad L_e = L - 2h = 22 - 2 \cdot 3 = 16\text{mm}$$

$$A = t \times 2\Sigma = 67.2\text{mm}^2 \quad F_{\text{Shear}} = S_{SY} \times A = 5376\text{N} \quad FOS = \frac{F_{\text{Shear}}}{F} = \frac{5376}{130} = 41.354$$

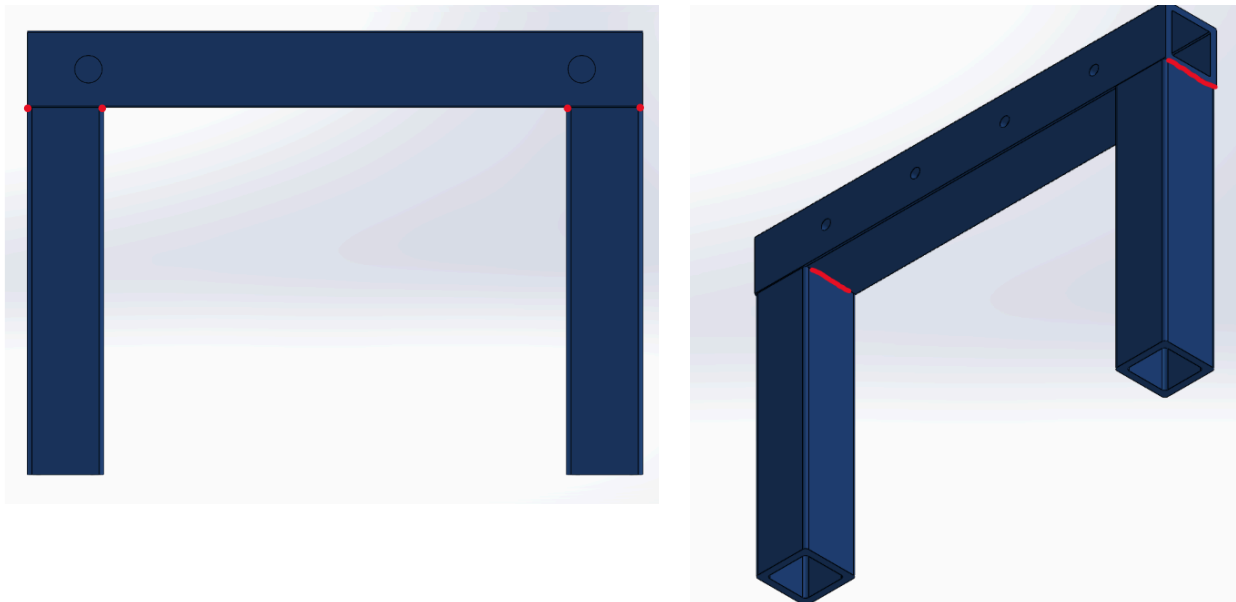


Figure 4.5.1.1: CAD section view of welded SHS

4.5.2 Fastener Selection

A range of fasteners were selected for use in this assembly for practical reasons. Fastening the grate, polycarbonate screen, and SHS sections to the face plates was done with socket cap head bolts screwed into tapped holes in the faceplate. Alternatively, fastening of the L brackets and grate to the SHS sections is accomplished with the use of rivet nuts, a special threaded rivet which is used in place of a tapped hole.

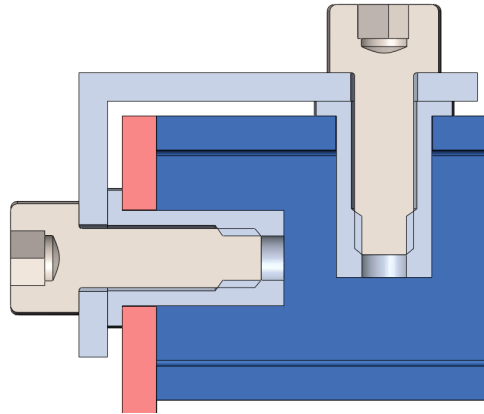


Figure 4.5.2.1: CAD section view of rivet nuts used to secure the L-brackets to the SHS and face plates

The teeth are assembled in such a way that they are easily disassembled for maintenance and replacing if needed. The teeth, and their spaces which provide clearance for the spinning blades, have clearance holes for a single M10 threaded rod which slides through the entire rack and is secured with nuts onto either side of the grinder. The teeth are only translationally constrained by the rod, but are free to swing. The rotation of the teeth and teeth spacers is prevented by the square hollow sections above and below.

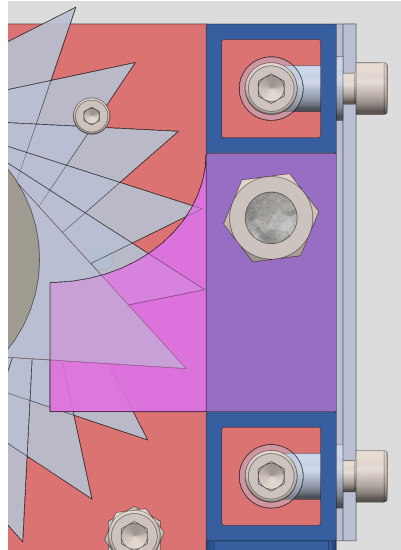


Figure 4.5.2.2: CAD section view showing how teeth and teeth spacers are restrained rotationally by the SHS

This was chosen to improve the serviceability of the teeth and reduce the overall cost by replacing an otherwise custom machined shaft/housing for the teeth.

4.5.3 Grate Construction

The grate will be assembled from a perforated aluminium sheet and three pieces of aluminium sheet metal cut to the appropriate size, and for the perforated sheet, bent to the correct geometry. These 4 pieces of Aluminum would then be set up in a jig, with a weld bead segments added to the main edges of the perforated sheet and tack welds connecting the edge pieces together. The load case of the grate is not expected to be high but tack welds would be insufficient around the perforated sheet as poor placement could significantly impact the performance of the welds.

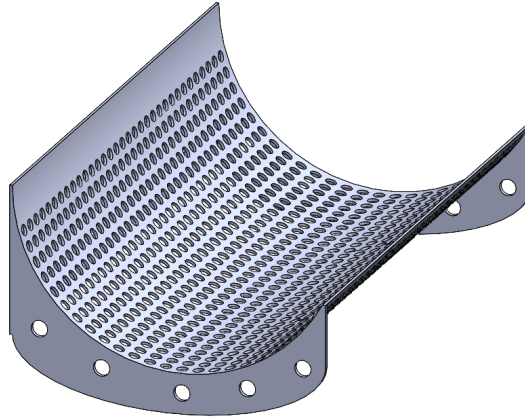


Figure 4.5.3.1: CAD of the isolated grate

The grate will be secured with 4 permanent rivets across the back, holding it to the SHS frame and the window. Each side is secured to the respective panel with 5 M5 nuts and bolts. This is a very sturdy combination of boundary conditions but it is important to note that one side is free, and that the loading is somewhat ambiguous. In the worst case there could be multiple pieces of plastic stuck in the grate being sheared by the blades at once leading to a larger loading.

4.6 Development of reinserting pieces larger than 5mm

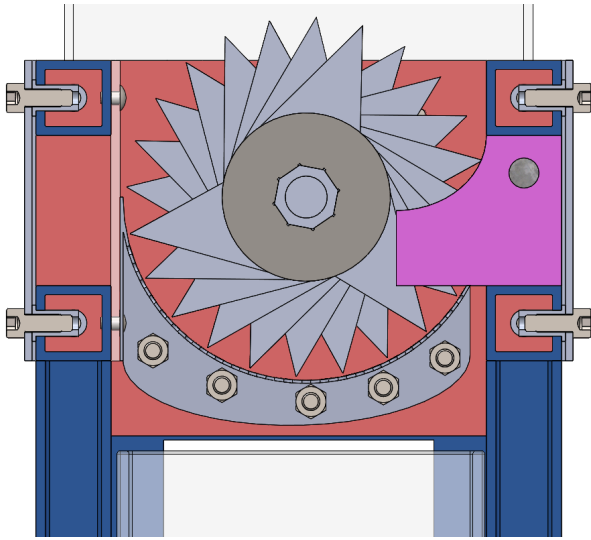


Figure 4.6.1: Cross section view of shredder showing how the blades interact with the grate.

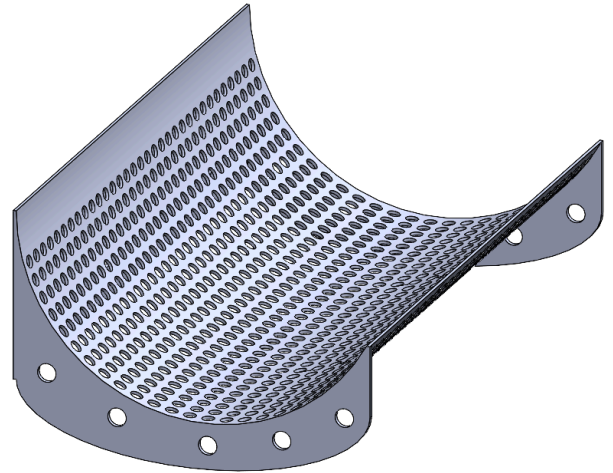


Figure 4.6.2: Isometric view of the grate used in the shredder.

Ensuring all plastic that is put into the shredder is reduced to no greater than 5mm in any dimension is a critical consideration of the design. To facilitate this the grate was developed, the holes in the grate are 4mm in diameter, this is to reduce the chance of larger pieces falling through. The grate is designed to curve with the blades as can be seen in figure 4.6.1, allowing larger pieces to be automatically reinserted into the shredder. The grate is secured to the frame with fasteners on three sides, the fourth side is positioned under the cutting teeth to immediately collect plastic that falls through.

The holes of the grate are also a secondary cutting mechanism, as the larger plastics get stuck the blades can dislodge or cut those chunks. This component is intended to have a smooth surface to reduce maintenance requirements and ensure that plastics are not prevented from going through the grate.

The grate was unaffected by changes to the blades and other components as the outer diameter of the blades remained unchanged and there were no notable issues with the design. The blade design increases the cutting area of the blade and would ensure when one blade is in contact the adjacent blades are not, reducing the likelihood of jamming. The teeth remain curved to lower the torque requirement and distribute the cut over a slightly larger part of the rotation, also helping to mitigate jamming of the shredder. The blades and teeth remain at a spacing of 3.2mm, this also goes towards reducing the plastic size to less than 5mm in any dimension. A design that combined coarse and fine blades was considered but the final version was chosen to keep the design simpler.

4.7 Analysis of Additional Components

4.7.1 Grinder Frame

The legs of the frame are constructed from pre-fabricated aluminium square hollow section, with welds and holes drilled for fasteners. The square hollow section was selected due to the low cost of a pre-existing length with a high moment of area to help improve resistance to buckling and bending, as well as providing a strong base under which a removable collection tub can sit.

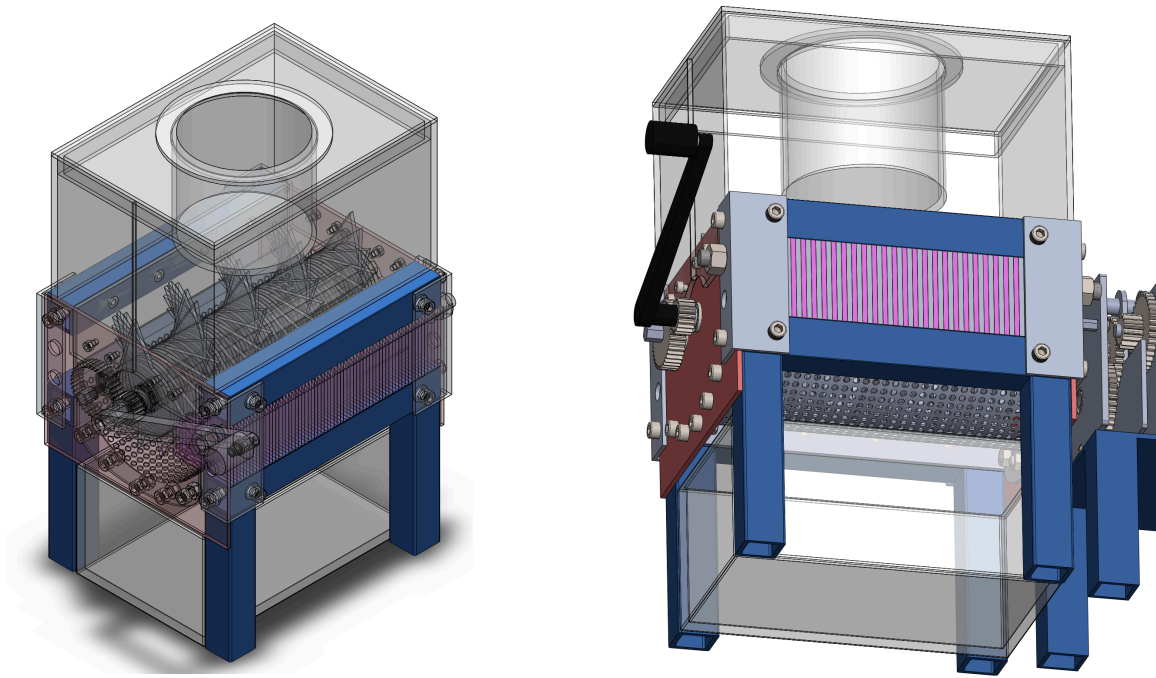


Figure 4.7.1.1: Grinder CAD with the square hollow section framing shown in blue

The ends of the grinder are constructed from waterjet cut aluminium plate, allowing again for low-cost manufacturing and simple assembly through clearance holes, tapped holes, and pressed bearings.

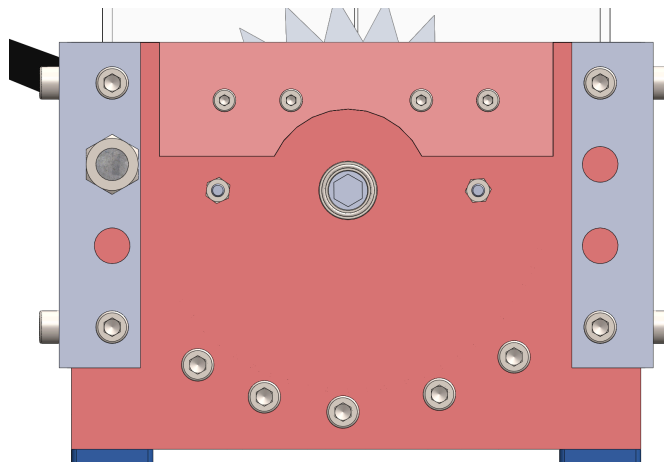


Figure 4.7.1.2: CAD of the motor-side face plate, showing the various fasteners utilised.

4.7.2 Grate

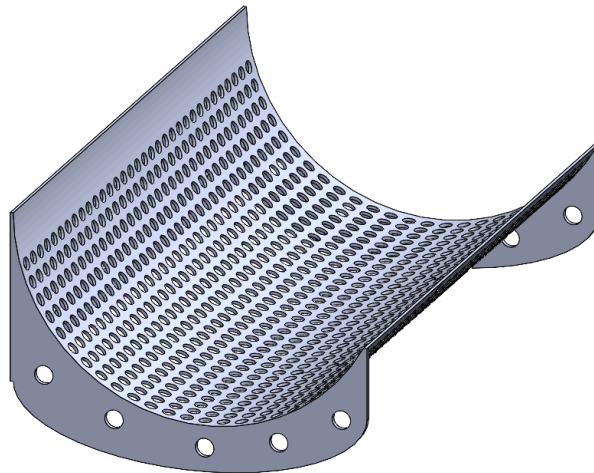
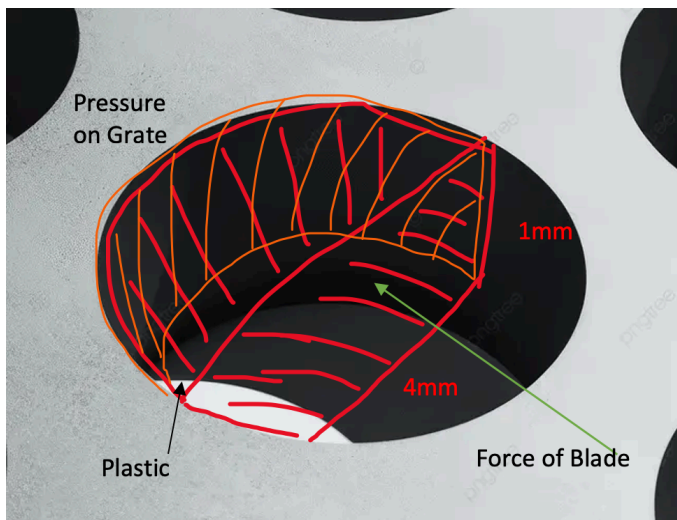


Figure 4.7.2.1: CAD Isometric view of the grate used in the shredder

The Boundary conditions of the grate are unclear as there are 14 integration points on the grate. Due to this only the perforated will be analysed in this section for yield failure.



$$A_{Plastic} = 6.28mm^2$$

$$\sigma_{yield\ Plastic} = 60MPa$$

$$F_{Blade} = \sigma_{yield\ Plastic} \times A_{Plastic} = 376.8N$$

$$A_{Hole} = 1 \times 4 = 4mm^2$$

$$P_{Grate} = \frac{F_{Blade}}{A_{Hole}} = 94.2MPa$$

$$\sigma_{yield\ 7075} = 470MPa$$

$$SF = \frac{\sigma_{yield\ 7075}}{P_{Grate}} = 4.98$$

Figure 4.7.2.2: CAD close-up illustrating the interaction between the plastic and the grate holes

The Force of the Blade is the force required to shear off the plastic that is stuck in the hole. That force is then applied over the area of the hole wall giving the pressure. The factor of safety is very high but it is important to consider that while this is a worst case load, there will be other forces acting on the grate at other sites. Also this assumes that an uniform material, but the perforated metal would be weaker than its solid counterpart. Due to the uncertainties of the simulation the factor of safety of 5 is adequate for this component.

4.8 Tolerance Analysis

The following section of the report covers tolerances for some of the most essential components of the shredder as a whole. All fit sizes have been found using ISO 286-2 Standards, and will be written in the form of the unilateral approach, whereas all parts are at Maximum Material Condition (MMC). Each tolerance has been chosen with the knowledge that any incredible small tolerances could drastically increase the cost of the design, especially on pieces that are produced at mass, such as the blades and the spacers.

4.8.1 Connection of the Shaft to the Bearing Inner Ring

Ø12₀^{+0.009}
(M)

With the assumption that the bearings have an exact inner diameter of 12mm, the cylindrical component of the shaft that connects to the inner diameter of the bearing must cause either a transition fit or a light interference fit. This type of fit prevents the shaft from slipping inside the bearing, causing any unnecessary wear or fatigue on the shaft. Hence, the selected fit type was a k5 fit. At a diameter of 12mm, this has a lower end tolerance of 1µm and a higher end tolerance of 9µm, with both of these tolerances also equating to the minimum and maximum amounts of interference the shaft and bearing can experience. This light interference fit will mean that the bearing can not just slide on the shaft to connect them to one another, there will need to be the use of a machine to press on the bearing, and potentially the use of heating to allow them to initially connect to one another. This type of fit will mean that the bearing will not simply slide off of the shaft, which is an intentional outcome that increases the operation efficiency of the system, however, this does mean that methods used to combine the two components may also be used to separate the parts from each other in a circumstance where a blade needs to be replaced.

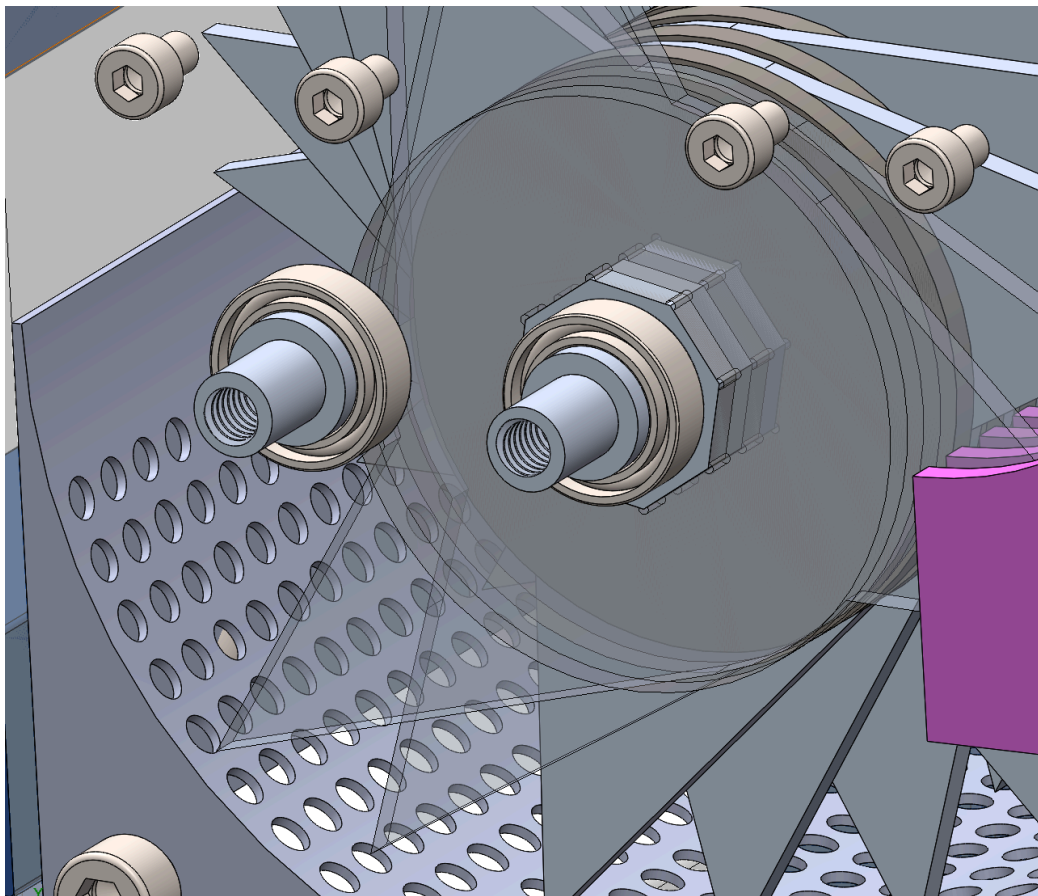


Figure 4.8.1.1: CAD of the interaction of the Roller Bearings with the Shaft.

4.8.2 Connection of the Bearing Outer Ring to the Frame of the Design

Ø17
_992
M

With the continuation of the assumption that the manufactured bearings are perfectly set to size and have an outer radius of 18mm, the hole in the frames which the bearings connect to must also be around the space of a transition fit. This type of fit prevents the bearing from knocking or rattling on the frame in a circumstance when a shock load is experienced inside of the shredder. The selected hole fit type for this circumstance is J7, which has a lower end tolerance of $-8\mu\text{m}$ and a higher end tolerance of $10\mu\text{m}$, and this tolerance applies to all three holes required on the side of the frame with alternating gears, as the bearings inside those gears are also all the same size.

This type of fit means there is a maximum interference between the bearing and the hole of $8\mu\text{m}$, and a maximum clearance of $10\mu\text{m}$. With the possibility of a slight interference, the bearing may need to be lightly tapped into the holes on the frame, with the use of something small like a plastic mallet, still making the component removed relatively easily for maintenance on the device, whilst still remaining stable during the shredder's operation.

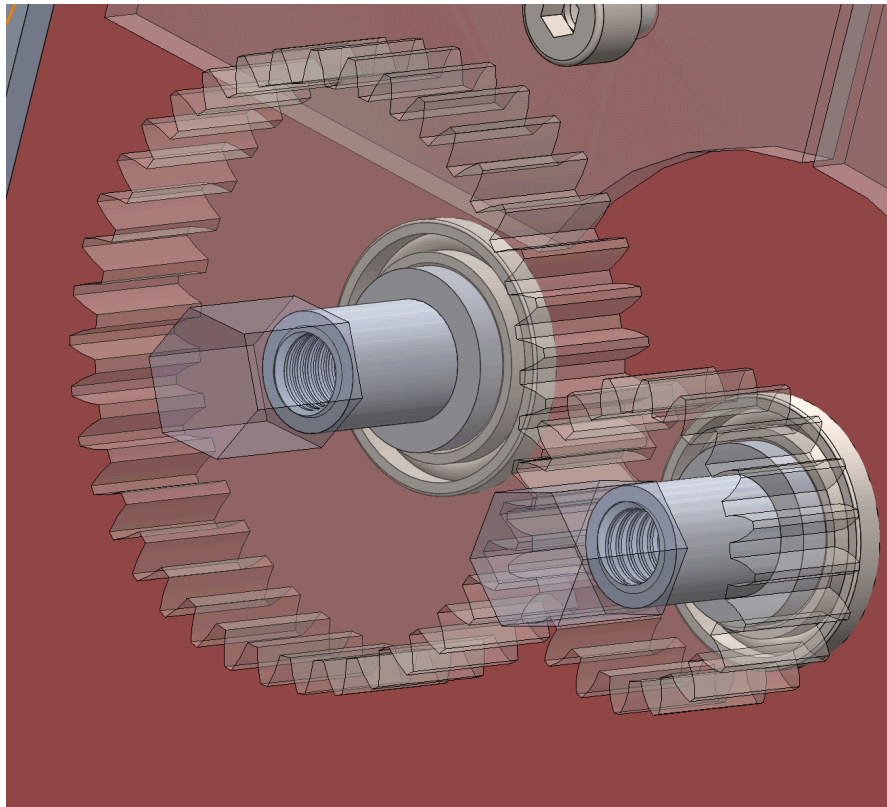


Figure 4.8.2.1: CAD of the interaction of the Roller Bearings with the Frame.

4.8.3 Connection of the Shaft to the Blades and the Spacers

20.0 $^{+0.033}_{-0}$ (Hole)
 19.9 $^{+0.007}_{-0.013}$ (Shaft)
 93M

19.9 $^{+0.007}_{-0.013}$ (Shaft)
 93M

The section of the shaft that connects to holes in both the blades and the spacers is an octagonal shape, so rather than using a measurement of diameter as the measurement of focus, the focus will rather be on the measurement across flats, meaning the distance from one flat face of the octagon to the flat face acting parallel to it. In this particular case, the octagonal distance across flats begins at a standard of 20mm. These particular components need to fit snug to one another, but would hover around a clearance to transition fit, because of the sheer number of blades and spacers needing to be added to the shaft, making an interference fit unfeasible. Any slight clearance between the hole and the shaft will not matter when all blades and spacer are tightened to be pressed against one another. For this component, the hole fit of H8 was chosen, with a g6 fit being chosen for the shaft. At 20mm, the H8 hole tolerance has a lower end of 0 μ m and a higher end of 33 μ m, whereas the g6 tolerance on the shaft has a lower end of -20 μ m and a higher end of -7 μ m. When these tolerances are taken into account, it gives the shafts interaction with the blades and the spacers a minimum clearance of 7 μ m and a maximum clearance of 53 μ m.

Additional GD&T measurements are essential for this component of the design; there needs to be a rough guide for how central the octagonal hole is on the blades and the spacer. They will be given a position control, on the hole, of 0.1mm, as well as a flatness control of 0.05mm. Finally, a 0.02mm position control on the shaft would assure that there is an accurate octagonal shape being produced that can easily interact with the holes in the blades and spacers.

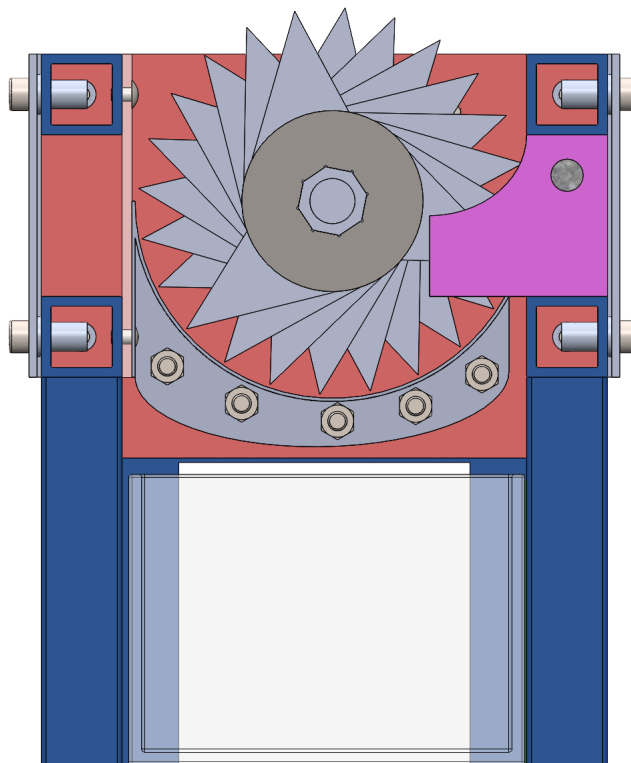


Figure 4.8.3.1: Section view of the inside of the Shaft.

4.8.4 Clearance Between Blades and Spacers

3.00 (Blade
(M) and
Teeth)

3.20 (Spacer
(M))

The CAD model initially had the spacers being 0.2mm thicker than the blades, alternating between the two as the thickness of each blade and spacer combined to a length of 201.6mm ($33 \times 3.2 + 32 \times 3.0$). To avoid creating a circumstance where the blades and spacers are connected to each other and are longer than this length, the MMC for the blades and the spacers must be 3.0mm and 3.2mm respectively. Each component will be given a negative tolerance range of $100\mu\text{m}$, which means that if the assumption is made that the blades and the teeth have the same tolerances, there will be a minimum clearance of $100\mu\text{m}$ between the teeth and the spacers, with the maximum being a 300 μm clearance. In a scenario where every component is at Least Material Condition (LMC), the stack length of the shaft becomes 195.10mm ($33 \times 3.1 + 32 \times 2.9$), meaning the stack length of the shaft can vary by up to 6.5mm. If the stack length happened to be the shortest possible, the use of nuts, bolts or shims would be needed to fill the empty space this gap would leave.

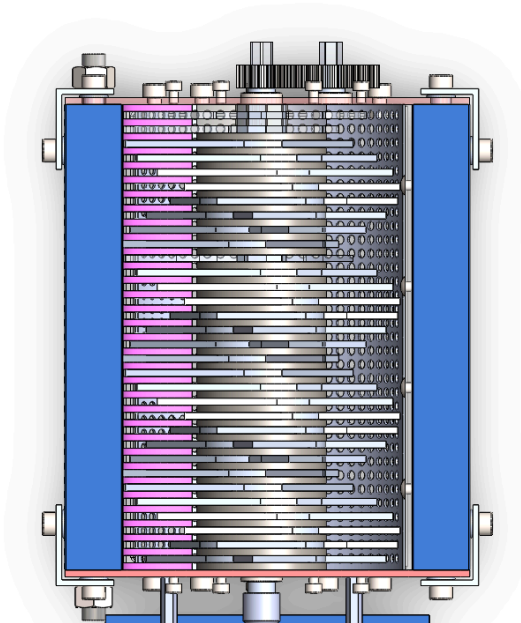


Figure 4.8.4.1: Top-down view of the Spacers and Blades along the Shaft.

4.8.5 Connection of the Handle and the Shaft

8.66 (Handle
Hole)

8.64 (Gear
Shafts)

Designing the tolerance for the connection of the handle to the shaft is slightly different to when other components are connecting to the shaft. There needs to be a larger amount of clearance between these two components, as the handle needs to be capable of sliding off of the shaft so that it can alternate between the three different sets of gears with relative ease and without the use of external equipment. While this does have the unfortunate consequence of causing a slight amount of wiggling in the handle as someone goes to operate the shredder, it is a necessary feature of the design. The connection point of the handle to the shaft is a hexagonal shape, and so the measurement of focus will once again be across flats, as it was for the connection of the shaft to the blades and the spacers. The chosen fit for this design is a H8f7 fit, which is a slight clearance fit. With the across flat measurement of the octagon being 8.66mm, the H8 tolerance has a tolerance low end of 0 μ m and a high end of 22 μ m, whilst the f7 tolerance on each of the gear shafts has a low end tolerance of -28 μ m and a high end of -13 μ m. This means that there will be a minimum clearance of 13 μ m in the design, and a maximum of 50 μ m.

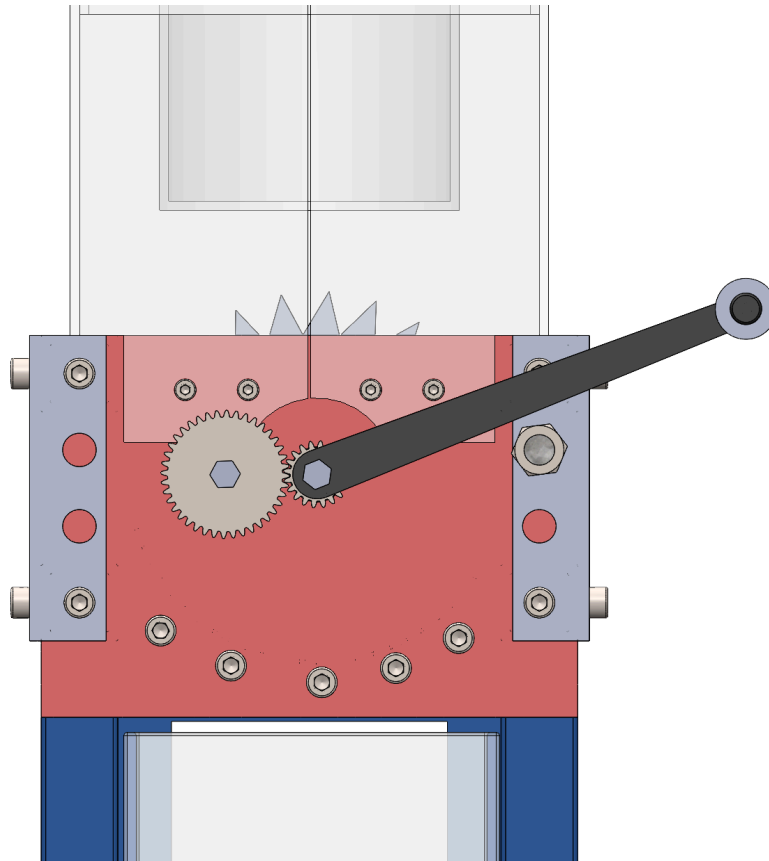


Figure 4.8.5.1: Front-on view of the Handle connecting to the Shaft.

4.9 FEA Analyses

4.9.1 Shaft Convergence report

Finite Element Analysis (FEA) was performed in SolidWorks Simulation to evaluate the shaft's stress distribution, deformation, and fatigue life. A curvature-based tetrahedral mesh was applied, with refinement in critical regions such as the hexagonal blade interface, fillets, and bearings. The mesh parameters adjusted during optimisation included the minimum element size, maximum element size, minimum number of elements in a circle, and element size growth ratio, allowing accurate capture of geometry while maintaining computational efficiency.

A mesh convergence study was conducted by systematically refining the mesh — primarily by reducing the minimum element size — and recording the resulting peak von Mises stress. Convergence was achieved when the difference in peak stress between successive refinements was below 5%, indicating that further mesh refinement would not significantly affect results. The final mesh settings used a maximum element size of 3 mm, minimum element size of 0.3 mm, growth ratio of 1.3, and at least 12 elements per circular edge, producing a high-quality mesh suitable for static and fatigue studies.

Table 4.9.1.1 Mesh Parameters for shaft FEA

Parameter	Final Setting	Description
Maximum Element Size	3.0 mm	Controls global mesh coarseness
Minimum Element Size	0.3 mm	Controls refinement near high-stress regions
Minimum Elements in Circle	12	Ensures accuracy in curved shaft sections
Element Size Growth Ratio	1.3	Gradual size transition to improve accuracy
Convergence Criterion	<5%	Verified by comparing peak von Mises stress values

Table 4.9.1.2 Mesh parameters and corresponding stress and percentage change

Mesh Trial	Min. Element Size (mm)	Max. Element Size (mm)	Elements per Circle	von Mises Stress (MPa)	% Change
1	1.0	5.0	8	2860	—
2	0.6	4.0	10	3030	5.9%
3	0.4	3.5	11	3100	2.3%
4	0.3	3.0	12	3126	0.8%

Under nominal loading conditions, the shaft experienced a peak von Mises stress of 3.126×10^3 MPa, concentrated at the hexagonal blade interface where torque transfer occurs.

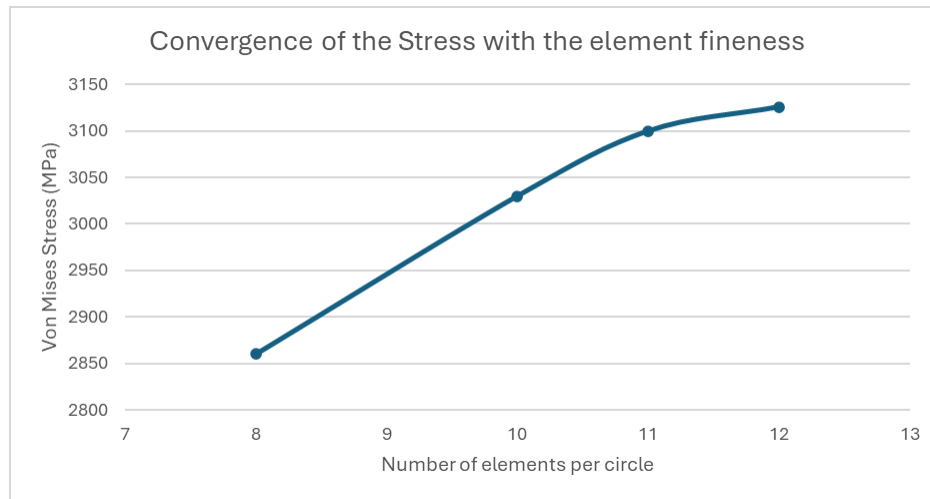


Figure 4.9.1.1: FEA convergence plot shows acceptable convergence after 4 iterations

4.9.2 Optimisation of the Shaft

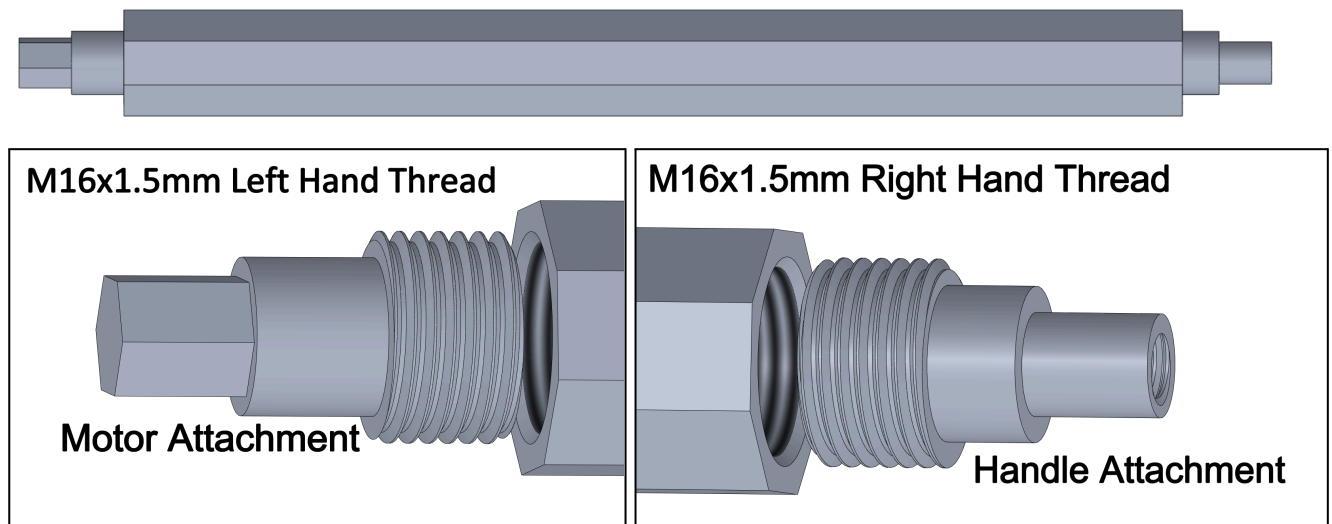


Figure 4.9.2.1: CAD of the shaft after optimisation, showing the threaded closures for attachment

The design of the main shaft presented in the preliminary submission was primarily focused on function rather than manufacturing feasibility, which initially resulted in higher material costs and unnecessary structural complexity. The original configuration did not account for the cost implications of producing a solid shaft with fully machined ends, leading to an over-designed component with excess mass and unutilised material. Furthermore, early assumptions about the shaft diameter led to an overly conservative design with a high factor of safety, which could have been reduced through more detailed analysis. In response to these findings, several design refinements were implemented to reduce cost, simplify manufacturing, and optimise structural performance while ensuring that the shaft could still safely withstand the applied torque and bending loads.

To optimize the design, the priority was to create a shaft that remained structurally sufficient to transfer torque between the motor and the blades while reducing fatigue and deflection through geometric and material refinement. The first major change was to adopt a hollow configuration, which reduced weight and improved the stiffness-to-mass ratio. The final design incorporated an outer diameter of 20 mm, corresponding to the inner diameter of the octagonal blade interface, and a wall thickness optimised through theoretical and FEA validation.

		Current	Initial	Optimal (15)	Scenario 8	Scenario 9
Inner Diameter		18mm	18mm	16mm	9mm	10mm
Minimum Factor of Safety ²	> 1.200000	6.613e-01	6.613e-01	1.261e+00	6.114e-01	6.649e-01
Stress1	Monitor Only	7.637e+08 N/m ²	7.637e+08 N/m ²	4.005e+08 N/m ²	8.259e+08 N/m ²	7.595e+08 N/m ²
Total Cost1	Minimize	0.0881912	0.0881912	0.0881912	0.0881912	0.0881912

Figure 4.9.2.2: Design optimisation table from the SolidWorks Design Study

A hexagonal connection interface was chosen for the blade attachment, allowing each blade to fit securely onto the shaft like a jigsaw puzzle piece. This configuration provided accurate alignment and simple assembly while slightly increasing stress concentrations at the corners. However, finite element analysis (FEA) confirmed that the maximum stress remained within acceptable limits at 75.48 kN/m², with a factor of safety (FOS) of 1.27 under the maximum expected torque and bending moment. The graphs show that the component experiences stress of 9885 N/m² for majority of the shaft, and only experiences von mis stresses above the yield strength (5.05×10^8) where and when the motor is operational

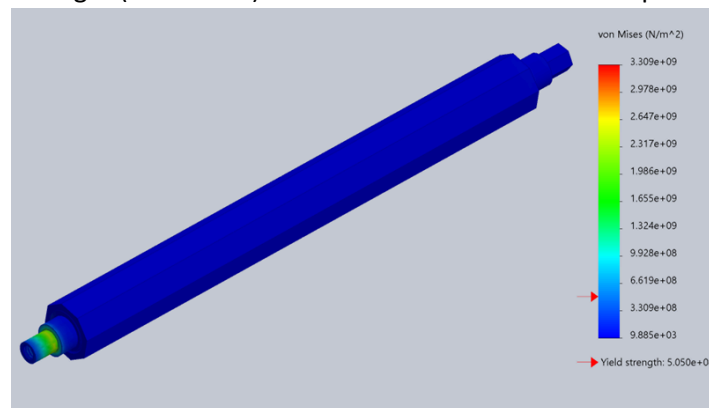


Figure 4.9.2.3: Von Mises stress plot and legend

The endurance limit confirmed the shaft's suitability for cyclic loading over approximately 4×10^7 cycles, well above the operational requirement. The fatigue damage will be largely experienced through the end pieces, rather than the shaft itself.

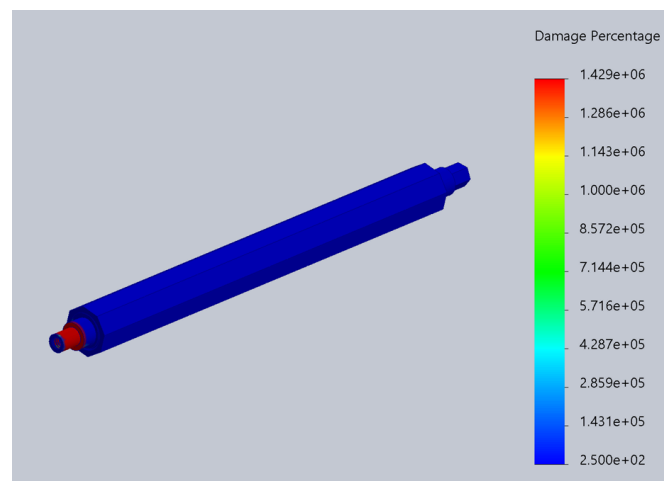


Figure 4.9.2.4: Fatigue damage plot

4.9.3 Optimization of the Blades



Figure 4.9.3.1: Design of blades in the Preliminary design

The design of the blades that was presented in the preliminary submission failed to consider the cost that would be associated with having a design with such round edges and hence a more complicated geometry. As a result of attempting to analyse the geometry of the blades, various assumptions were made about the contact surface of the blades on the plastic, which resulted in a significant factor of safety that was unnecessary for the design. Any slight changes that could have been made about the blades were imperative to cutting costs in the design, with the knowledge that over 1500 of the blades would be required to create 50 prototypes of the shredder.

To optimize the design, the priority was to create a blade that was still more than sufficient to handle the 162N of force required to break through a piece of plastic, while also being cost efficient and simplistic in its geometry for machining purposes. The first change that was made occurred before any FEA analysis took place, and that was a change in material. Originally, the blades were made of a 7075-T6 Aluminium alloy, but to preserve costs, the blades were changed to now be made of a 1060 Alloy instead, a material which is cheaper by volume.

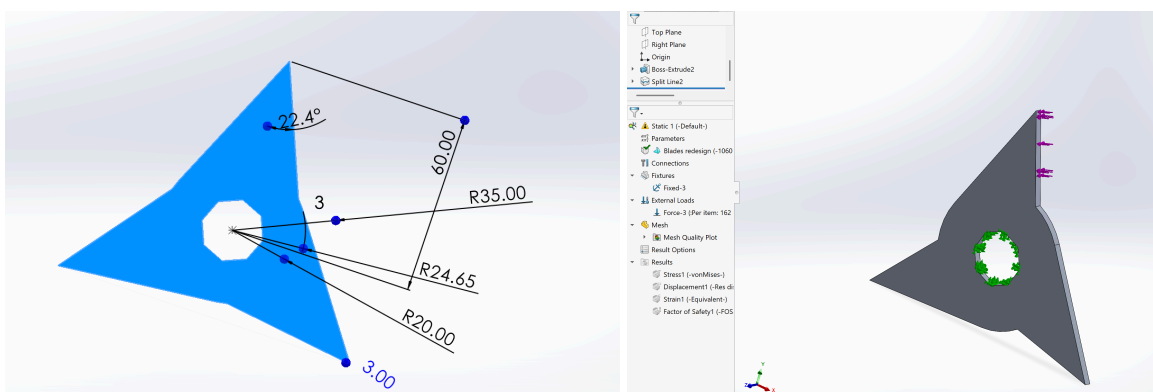


Figure 4.9.3.2: Optimization Measurements of Newly Designed Blades and the Static Study Set-up

After a static study was run on SolidWorks that involved one edge of the blades experiencing all of the 162N of force as the octagonal hole was set at a fixed geometry, measurements had to be selected to be optimised. The two components that were chosen were the radius of the circular component of the blades (seen in Figure above as the measurement of R24.65) and the angle of attack of the blades (seen in Figure above as measurement 22.4°). A design study was then set up with the range of the radius being between 20mm and 35mm, while the range of the angle of attack was set between 5° and 25°. These values were a result of seeing how different features of the blades intersected at the extreme ends of each case. The design study was set with a constraint of producing a minimum factor of safety of 2 over the blade, whilst also having the goal of minimizing the mass of the blade.

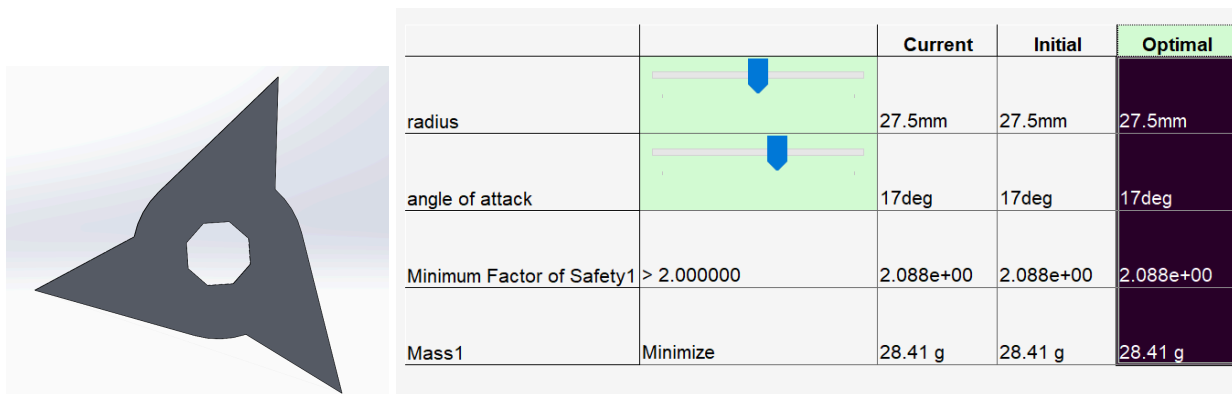


Figure 4.9.3.3: Appearance of Optimized Blades alongside Design Study parameters.

The design study produced an optimal blade which has a radius of 27.5mm and an attack angle of 17°. The minimum FOS of 2.088 exceeded the minimum of 2 that was required, and came to a weight of 28.41g.

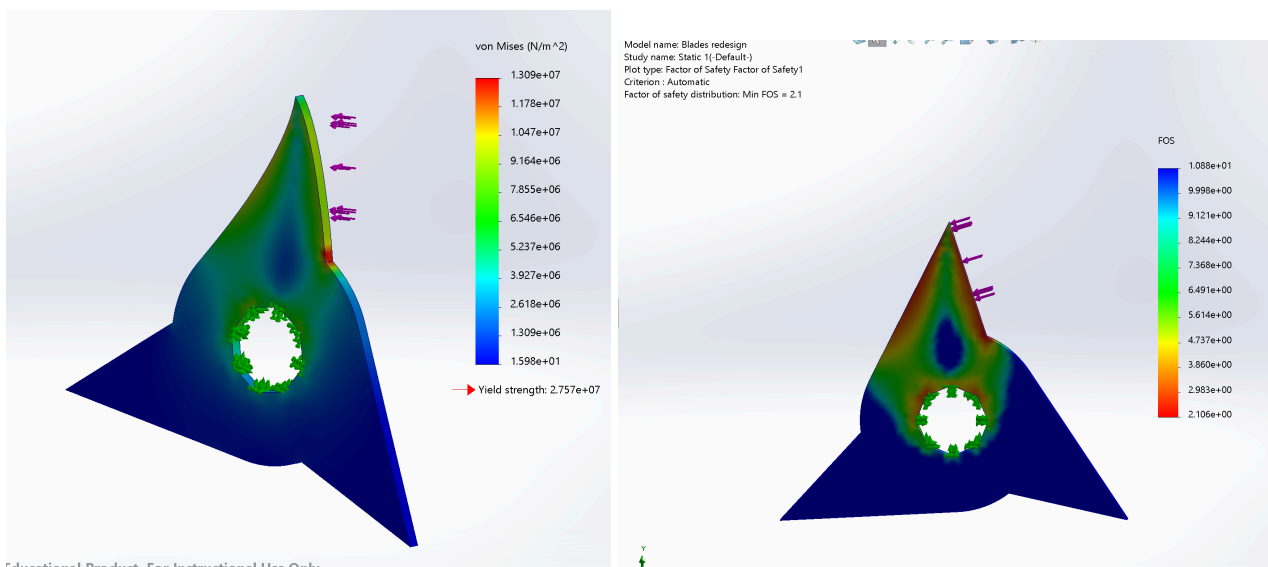


Figure 4.9.3.4: Results of the Static Study with Optimised Dimensions

5 Performance and Sustainability

5.1 Performance and Cost Summary

The shredder is designed with sufficient safety factors to ensure reliability withstands the operational stresses. It is designed to process 6 bottles per minute at 60 rpm which allows for 10 rotations per bottle with three cutting blades per rotation. The cutting face is 25mm long from the tip of the blade as calculated by measuring the distance the blade is in contact with the tooth on Solidworks. Therefore the effective cutting length in a minute is $25 \times 3 \times 60 = 4500\text{mm}$. As stated in the brief the 6 bottles are 375ml which are significantly smaller than a 500ml bottle, found to be roughly 200mm tall². The longest time for a bottle to be shredded would be when it is placed vertically so we assume the worst case and all 6 enter vertically. The measurements of 375ml bottles were elusive and vary greatly based on geometry so let's assume the height of the 500ml bottles as a conservative approach. The 6 bottles at 200mm height have a total combined height of 1200mm which is just over a quarter of the effective cutting length so the shredder will confidently shred more than 6 bottles per minute.

In order to guarantee prolonged operating capacity 1060 aluminium was selected for the blades for its rust resistance, as rust could pose an issue for steel blades due to potential water left in the plastic bottles. Stainless steel is an alternative option, however that significantly increases the cost of the blades which contribute a large portion of the total spending. Spending needed to be reduced to fit the \$500 prototype budget, hence in optimisation minimising cost was important. The total cost amounted to \$479.99 AUD and Appendix B features a detailed breakdown of each component and sub assemblies.

2 - Single Use plastic bottle on Dimensions.com

5.2 Sustainability Analysis

A Life Cycle Assessment (LCA) was conducted in SolidWorks using the CML method to evaluate the shaft assemblies, as well as the blades' environmental impact over their full 5-year lifecycle, including material production, manufacturing, use, and disposal. The manufacturing region was set to Asia, reflecting a common supply chain model, while the primary use region was set to Australia.

5.2.1 Sustainability of the Shaft

The shaft is manufactured via machining processes from a single billet of material. To reduce material use and weight, the central section of the main shaft is hollowed out. This design strategy directly minimizes the raw material required, thereby lowering the embodied energy and carbon footprint associated with the material phase. Following the hollowing process, the necessary threads are machined onto the shaft ends. The assembly is designed in three separate parts to accommodate its function within the shredder, allowing the mechanical motor and handle to effectively contribute torque. This modular design could also facilitate repair or replacement of individual components, potentially extending the product's service life beyond the assessed 5-year period.

Life Cycle Impact Analysis (5-Year Scenario) shows that environmental impacts are heavily concentrated in the manufacturing phase, with the material production phase being the next most significant source.

Table 5.2.1 Total Environmental Impact of Shaft over 5-Year Lifecycle

Impact Category	Total Impact	Manufacturing Contribution	Material Contribution
Carbon Footprint	210 kg CO ₂ e	200 kg CO ₂ e (95%)	8.2 kg CO ₂ e (4%)
Total Energy Consumed	3000 MJ	2900 MJ (97%)	100 MJ (3%)
Air Acidification	1.4 kg SO ₂ e	1.3 kg SO ₂ e (93%)	0.060 kg SO ₂ e (4%)
Water Eutrophication	0.051 kg PO ₄ e	0.049 kg PO ₄ e (96%)	2.0E-3 kg PO ₄ e (4%)

As shown, over 90% of the environmental impact arises from the manufacturing process, highlighting machining's energy-intensive nature as the primary contributor to the shaft's footprint. In contrast, impacts from transportation (0.013 kg CO₂e) and end-of-life (0.243 kg CO₂e) are minimal. The material cost is approximately \$1.40 USD (\$2.14 AUD). The product consumes no operational energy, reflecting its long service life, and at end-of-life, it is assumed that 50% is recycled, 40% landfilled, and 10% incinerated.

At the component level, the main shaft (Shaft_v5) dominates the impact, contributing about 35 kg CO₂e and 500 MJ of embodied energy, while the end fittings add smaller but notable portions. This supports the design decision to hollow the primary shaft, effectively reducing material use and total emissions.

In conclusion, improving sustainability depends on reducing the energy intensity of manufacturing, as material-related impacts have already been mitigated through lightweight design. Over its 5-year lifespan, the shaft provides a durable, low-energy solution with a significantly lower environmental cost than disposable alternatives.

5.2.1 Sustainability of the Blades

The blades are manufactured via machining processes from a single billet of material (Aluminium 1060).

Life Cycle Impact Analysis (5-Year Scenario) shows that environmental impacts are heavily concentrated in the manufacturing phase, with the material production phase being the next most significant source. It must be considered that these values in the following table are the impacts of a singular blade, when the shredder design contains 32 blades.

Figure 5.2.2 Total Environmental Impact of a Singular Blade over 5-Year Lifecycle

Impact Category	Total Impact	Manufacturing Contribution	Material Contribution
Carbon Footprint	14.4 kg CO ₂ e	14 kg CO ₂ e (97%)	0.389 kg CO ₂ e (3%)
Total Energy Consumed	145 MJ	4.8 MJ (96.5%)	100 MJ (3.5%)
Air Acidification	0.194 kg SO ₂ e	0.191 kg SO ₂ e (98.6%)	0.0026 kg SO ₂ e (1.36%)
Water Eutrophication	0.0075 kg PO ₄ e	0.0074 kg PO ₄ e (99%)	0.00083 kg PO ₄ e (1%)

As shown, and as was the case with the analysis of the shaft, over 90% of the environmental impact arises from the manufacturing process, once again highlighting machining's energy-intensive nature as the primary contributor to the environmental footprint of all components of the shredder. Additionally, impacts from transportation (6.1E-4 kg CO₂e) and end-of-life (2.3E-4 kg CO₂e) are also considered small, even though it is approximated that the material will have to travel around 8700km to be used. The product also consumes no operational energy, reflecting its long service life, and at end-of-life, it is assumed that 30% is recycled, 70% landfilled, and 0% incinerated.

Throughout design iterations of the blades, shifting from a design that is more linear, rather than the curved blades seen in the preliminary design, not only reduced the cost of manufacturing the blades, but would also decrease the energy required to manufacture them as well.

Whilst these approximations have been made with a 5-Year Lifecycle in mind, calculations regarding endurance limits surrounding the blades suggest that the shredder could easily exceed this lifespan, meaning new blades will not have to be created every 5 years and hence produce a greater impact onto the environment.

5.3 Maintenance and Safety Features

5.3.1 Features Designed Especially for Maintenance

A priority of the design is that it is capable of being maintained over the span of a couple decades. Whilst every component that experiences high loads has been designed to have a high enough factor of safety to prevent theoretical failure, external variables can always affect the functionality of any mechanism. For this specific reason, the design has been created in such a way that replacing individual components which become faulty over time should be able to be done with relative ease.

Bearing Selection - Every bearing that connects to the red frame of the design has been given a type of transition fit that allows the bearing to be removed from the frame with the use of smaller tools. With the fasteners on the same face of the frame being able to be removed, the shaft in its entirety is capable of being removed from the design to allow maintenance to take place. This allows for not only the

maintenance of every component of the shaft, but also every component inside the frame that surrounds the shaft, that may not be as accessible when the shaft is present in the design, such as the grate or the teeth.

Clearance Fit on the Shaft - The clearance fit that is present between the shaft on to the holes in the blades and spacers allows for any one of over 60 pieces of the shaft to be individually analysed, or even replaced if necessary. The clearance fit allows for a snug fit for the components during operation, and does not limit functionality on the design if each piece is properly pressed together, but has the advantage of allowing each piece to slide on and off of the shaft easily enough that maintenance on the shaft is not a timely activity.

Visibility of the Shredder's Internals - The use of translucent Polycarbonate Externals on the design for components such as the lid, and a miniature window on the side of the design, allow for the visibility of the shredder's internal components as the machine operates. In the event of an unexpected failure, this visibility can serve crucial in identifying where the issue is, without having to fully take apart the device.

5.3.2 Features that Cover Safety Concerns

As is the case with any machinery as a whole, but especially machinery that is used for cutting material, there are a variety of safety concerns that must be addressed to allow the shredder to be used in an everyday family home whilst minimizing any potential dangers.

Limbs in Contact with Shredder - The large lid that has been placed above the blades of the shredder acts as a barrier from an individual's hand and the blades to prevent someone seriously harming themselves. The removable component of the lid is also large enough that bottles can be placed inside of the shredder, while remaining small enough to prevent someone from accidentally placing their hand inside of the device.

Shrapnel Produced by Shredder - As is intended from the design, the shredder will create small pieces of plastic from the bottles that it shreds. However, these small pieces of plastic are typically quite sharp and could cause harm to an individual in a multitude of ways. As mentioned previously, the lid serves as a barrier from the blades and the individual operating the machine, meaning that any pieces of plastic that may randomly spit upwards as the bottle breaks will not be able to fly into the operator's eye. Additionally, the use of the collection bucket at the bottom of the shredder, which can easily slide out of the device, grants the operator the opportunity of transporting the small plastic pieces to a heating device to be turned into printing filament, without ever having to touch the sharp pieces of plastic with their hands.

Interference of the Machine as the Motor is in Use - The removability of the handle between different gears ratios on the shaft also allows for the handle to be completely removed in a situation where the mechanical side of the shredder is in use, meaning the handle will not spin around on its own and provide a potential to anyone in close proximity to the shredder.

Weight of the Shredder - The shredder has been designed to be as lightweight as possible, meaning that 1-2 people would be capable of carrying all parts of the design that are shipped already connected to one another, before bringing all of the pieces together in the home to the desired location of assembly.

6 Conclusion

Overall, the development of the plastic grinder has been a success. The performance of the blades and teeth, alongside the power outputted from the geared handle and motor, provide a strong confidence in the grinder's ability to process the required 6 bottles a minute. Additionally, the entire grinder, including the motor, gearbox, and handle, fits comfortably inside the required dimensions.

The approach to design was centered around serviceability, robustness, reliability, and longevity. This led to components with good factors of safety, simple manufacturing techniques, common materials, and little to no fatigue damage.

It was a particular challenge to ensure that the rate of bottle processing was fast enough under a human-powered crank motion. This required a fine balance between the number of arms on each blade, the geometry of the handle, and the gear ratios to ensure enough torque and angular velocity of the shaft. Whilst there is a high confidence level around this performance metric, physical testing is the only reliable way to validate the requirement with certainty. However, better modelling could be achieved using a program such as Ansys Rocky to analyse the performance of the grinder.

Financially, the \$500 budget placed a tight constraint on design. Cost was always considered, and is shown through our selection of waterjet cut parts, reducing the number of set ups for any machined parts, selecting cheap materials where appropriate, and the use of prefabricated square hollow sections and angle extrusions to limit further manufacturing. Ultimately, the budget was upheld. However, in future, there should be further exploration of industrial sources for components. Economies of scale would result in reduced prices for manufacturing, and the use of more appropriate materials. For example, the threaded rod is currently sourced from Bunnings and is galvanised steel, which is an unrealistic source for mass production, and an over-engineered solution.

7 References

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8 Appendices

Appendix A: Purchased Component Catalogues

Appendix B: Cost Analysis Breakdown

Appendix C: Detailed and Assembly Drawing

Appendix D: Week 9 Progress Review

Note: All appendices have been submitted alongside this report.